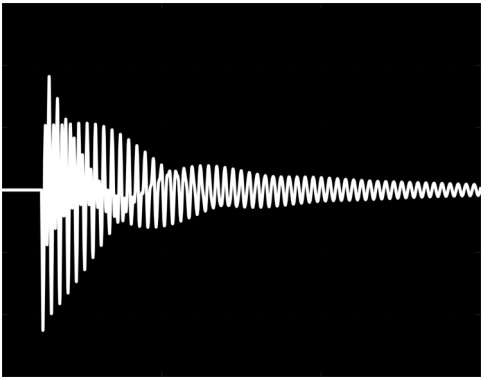
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# Reduced order model integrated control of beams



*A Thesis*

*Submitted by*

**PATEL HIRENKUMAR SANJAYKUMAR**

*For the award of the degree*

*Of*

**MASTER OF SCIENCE**

**(by Research)**

June 2021

# QUOTATIONS

*If you want to find the secrets of*
*the universe,*
*think in terms of energy, frequency*
*and vibration.*

NIKOLA TESLA

# DEDICATION

*To my beloved*

*Parents*

*and*

*Swapna*

# THESIS CERTIFICATE

This is to undertake that the thesis titled **REDUCED ORDER MODEL INTEGRATED CONTROL OF BEAMS**, submitted by me to the Indian Institute of Technology Madras, for the award of M.S., is a bona fide record of the research work done by me under the supervision of Dr. Shaikh Faruque Ali. The contents of this Thesis, in full or in parts, have not been submitted to any other Institute or University for the award of any degree or diploma.

**Place: Chennai 600 036**
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# LIST OF PUBLICATIONS

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# ABSTRACT

KEYWORDS: Flexible structrue; reduced order model; vibration control; finite element model; Euler Bernoulli beam; non-Gaussian random variable; linear quadratic regulator; Monte Carlo simulation.

Flexible structures are continuum modelled as infinite degrees of freedom. The finite element methods are used with a finite number of degrees of freedom to analyze these structures. For real time monitoring, analysis and control of the dynamics of these structures one needs simulations of a large degrees of freedom in real-time. This is computationally expensive. This study focuses on developing a reduced-order framework for dynamics and control of flexible structures. Essentially, in the proposed framework, full order state-space obtained from finite element modeling of the flexible structure has been reduced to a lower subspace using a reduced order algorithm, keeping the key dynamical characteristic intact. The transformation matrix for reduction has been calculated using the system equivalent reduction expansion process (SEREP). Traditionally, full order dynamical system is used to find the gain matrix to suppress the uncontrolled dynamics associated with the system. In this methodology, the reduced dimension of state space of the system is used for estimating the gain matrix using the optimal linear quadratic regulator (LQR). The gain obtained through the reduced system is subsequently used as a feedback to the attached actuators which then produce the required force to control the system. A numerical example of a flexible cantilever beam has been shown to investigate the effectiveness of the algorithm.

To understand the applicability of the proposed methodology, attempts are made to use the method for system with parametric uncertainty. The system parameters are mathematically modelled as non-Gaussian (Log-normal) random variables. The high fidelity finite element model is the starting point of the analysis. In this study, a reduced-order model is developed to reduce the system's state space dimension and further integrate with the control algorithm. Efficacy of the algorithm and probabilistic

characterization are ascertained through Monte Carlo simulation.

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# CHAPTER 1

# Introduction

## 1.1 Background

In recent years, self-sensing and automatic high-performance structures have been motivated by the various needs of modern automobile, aeronautical and space industries. This has led to tremendous advancement in the way engineers have designed structural systems. From elastic, inelastic, plastic designs, engineers have moved towards the development of novel smart intelligent systems. A structure with an implanted sensor unit, actuation unit and a control system that changes its shape and dynamic behaviour in response to any variation in the external environment can be termed as smart structure. These systems are made 'intelligent' to make decisions on their own under unforeseen environment and behave accordingly to reach the target set by the designers. The measured or sensed data is compared with the design/desired data, and any deviation from the desired response is reduced instantly using actuations. This means that the design engineer has to install a brain or a control algorithm in the loop to make the system behave automatically.

The design and analysis of intelligent structures involve the fusion of multiple disciplines such as materials science, information technology, sensing and actuating technologies, system and control theory, artificial intelligence and biomimetics. Examples of smart systems are microelectromechanical systems, autonomous unmanned aerial vehicles and morphing structures, to name a few (Bora *et al.*, 2017; Jang *et al.*, 2015). These structures use smart materials as actuating and sensing elements. To make structures automatic, a design engineer needs to understand its status quo every instant. This can be achieved using sensing of different means.

Materials with special properties such as changing shape when heated or under electric or magnetic field, producing electricity when compressed or heated, and changing their physical states when subject to a magnetic or electric field are examples of smart

materials (Leo, 2007). Some of these properties can be manipulated and used effectively in actuation and sensing. Integrating these smart materials into host structures gives them inherent self-sensing and self-correcting ability without any separate sensing or actuation unit.

One of the materials that can be used as an actuator and sensor is a piezoelectric material and have been successfully demonstrated by many researches (Chee *et al.*, 1998; Chorsi *et al.*, 2019). Piezoelectric materials generate an electric charge when subjected to mechanical deformation (direct piezoelectricity) and conversely produce mechanical strain under an applied electric field (converse piezoelectricity). The coupled electromechanical properties of the piezoelectric materials, high strain rates, simple mechanism of actuation, and their availability in different shapes and synthetic forms have made it possible to use them as one of the important actuation and sensing elements in structural control applications. Magnetostrictive material, shape-memory alloys, and magnetorheological fluids are some of the other smart materials in use today.

In addition, modern space, automotive, and aircraft structures need to be strong but lightweight. Composites, in which two or more different materials with different material properties and/or chemical properties are put together in some particular fashion and tailored to meet the required engineering properties, are promising for such applications. Due to their high stiffness to weight ratio, strength to weight ratio and ability to withstand high temperatures, composite materials are desirable for modern structural needs. By embedding piezoelectric elements into composite material structures, there are possibilities of creating high-performance flexible structures with high strength, high stiffness and lightweight with self-sensing and self-correcting ability. The next chapter will cover brief historical development of vibration control strategies for flexible structures.

## 1.2 Motivation

Bird and insect flights have always fascinated humans for many centuries, and the imitation of their flapping-wing flight has been the greatest aeronautical dream. Because of the aerodynamic and mechanical complexity, complete and exact analysis of the flap-

ping flight are not available. The main challenge to this problem lies with the mathematical description of these phenomena. Generally, these kinds of phenomenon are governed by PDEs and known as distributed parameter systems. To design structures that flap like birds, one needs to use a smart structure with precise control techniques. Some excellent research has been done in the past (refer to Chapter 2) which addresses this problem. Most of the authors have focused on employing increasingly complex control algorithms techniques over the numerically approximated model (FEA) to mimic the phenomenon. Only a handful of researchers tried to solve the problem by further reducing the dimensionality of the problem. This thesis has found this gap that is not being addressed in the literature to an extent. Thus this thesis seeks to apply the foundation of available theories in structural dynamics, reduced-order modelling and structural control to develop a unique methodology. This will test the feasibility of integrating the reduced order modelling technique to control the deterministic as well as stochastic structural systems. The developed methodology is evaluated by solving the linear cantilever beam structure incorporating piezoelectric transducers. Based on the research gap discussed, the next section provides the objectives of the thesis.

## 1.3 Objectives

The objectives of the thesis are as follows:

- Developing a finite element framework for beams integrated with macro fibre composites (piezoelectric composites) as sensors and actuators.
- Developing a reduced order method using SEREP to reduce the active DoF for the beam keeping the key dynamics intact. SEREP will also help in keeping the DoF associated with the sensing and actuation of the beam.
- Developing framework to integrate the RoM with the linear control technique with deterministic and stochastic parameters.

## 1.4 Scope of the research

Following are few limitations of the present study.

- Electrochemical hysteresis of the piezoelectric materials is not considered in the current study.

- The current study deals with the linear system dynamics and control.

## 1.5 Organization of the thesis

The flow of the thesis is as follows:

Chapter 1 which is this chapter has provided a background to the necessity of this thesis, followed by a motivation. The objectives and the scope of the thesis are drawn thereafter. The Chapter ends with a summary.

Chapter 2 documents the brief development in the field, control of distributed parameter systems. This chapter also discusses the past development in reduced order modelling of structures and control of them in both deterministic and stochastic sense followed by a summary.

Chapter 3 provides the brief overview of the smart materials. The mathematical model for piezoelectric transducer are also reported in this chapter. This chapter also discusses the mathematical modeling to understand and compare the behavior of piezo actuated bimorph structures. The mathematical modeling of piezo actuated bimorph beams is conducted using a distributed parameter Euler-Bernoulli beam theory. Predictions of the tip displacement are validated against the model developed in commercial FE simulation software (ABAQUS). The chapter concludes with a summary.

Chapter 4 discusses the philosophy and procedure to obtain the transformation matrix which maps the full order system to reduced dimension using SEREP. In addition to that this chapter also discusses the procedure to implement control algorithm on reduced order model. This chapter also includes numerical results obtained from solving a benchmark problem of cantilever beam under impulse and periodic loading followed by the summary of the chapter.

Chapter 5 discusses the underlying principle of the reduced order model (SEREP) for stochastically parametered system. Further, control algorithm for stochastic reduced order model is also derived in this chapter. The proposed mathematical framework is illustrated through a numerical example of stochastic cantilever beam under the impression of impulse and periodic loading at the end. The chapter winds up with a summary.

Chapter 6 consists of a summary of the thesis and concluding remarks on the present

study followed by a future scope.

## 1.6 Summary

This chapter describes the background of the problem and establishes the foundation of this thesis. This chapter also includes the motivation behind taking up this problem followed by deliverables and scope of the research.

# CHAPTER 2

# Review of literature

The preceding chapter provided a background, motivation and objective that shape this thesis. The nature of problem involves analysing systems that have distributed nature of dynamic properties like, mass, stiffness and damping. The final goal is associated with the control of vibration in these systems. The control logic is placed over reduced dimensions of these systems. This chapter, therefore reports a comprehensive review of literature pertaining to distributed parameter system (DPS) control and reduce order modelling of associated systems.

## 2.1 General review of literature on control of beams/PDEs

Structural control is primarily achieved using passive mechanisms (Smith, 2002; Rao, 2003), where a particular frequency is targeted for control. The controller becomes sub-optimal or inefficient at other operating frequencies. Furthermore, a passive mechanism is bulky and heavy in nature (Krenk and Hogsberg, 2014). Increase in application of lightweight flexible structures (Colorado *et al.*, 2012; Obradovic and Subbarao, 2011) has opened the scope of controller design of flexible system using active means (Ali and Padhi, 2009; Kugi and Schlachert, 2003; Ali and Ramaswamy, 2008; Hu and Ng, 2005; Ali and Ramaswamy, 2009). Distributed parameter systems (DPS) is an established research area in dynamics and control. It traces its roots back to the 1960s (Butkovskiy and Lerner, 1960). DPS are mathematically expressed using partial differential equations (PDEs) (Curtain, 2003). Although the overall objectives behind DPS's control remain the equivalent as those for lumped parameter systems like controllability, stability, optimality, etc, certain unique challenges consist of satisfying the spatial boundary conditions, collocated and non-collocated control design etc. As a result, many algorithms have been designed and implemented for the control of distributed parameter systems.

Wang and Tung were pioneers in this field and published their work (Wang and Tung, 1963), which laid the foundation for further development towards a unified control theory for DPS. The authors attempted to give a precise mathematical description of DPS in terms of a set of partial differential equations and discussed the concept of controllability and observability. The paper also describes the formulation of optimal control problem and optimality conditions for a broad class of DPS. A few years later, Wang used Lyapunov techniques (Wang, 1966) to address the problem of DPS stability with feedback controllers in the framework of PDEs.

Kim and Erzberger proposed a Hamilton–Jacobi approach to solve boundary control problem in the one-dimensional wave, using quadratic cost function (Kim and Erzberger, 1967). In addition, the authors developed a set of Riccati equations whose solution gives the closed-loop optimal control. Moreover, they also developed a solution technique for the Riccati equations based on the eigenfunction representation of the Green's function, which leads to the more straightforward solution demands solving only a set of ordinary differential equations.

Christofides and Daoutidis suggested a control design tool for a class of quasi-linear parabolic PDE driven systems (Christofides and Daoutidis, 1997). In which, the eigen spectrum of the spatial differential operator was partitioned into a finite-dimensional slow spectrum and an infinite-dimensional fast one that is stable. This methodology uses the Galerkin approach to develop a weak form of governing equation, then to a suitable ODE system for the PDE dynamics and design a nonlinear output feedback controller that guarantees stability while making the system follows the desired response.

Matsuno and Murata (Matsuno and Murata, 1999) published a paper that describes the development of a direct output feedback control law based on proportional, derivative and strain feedback (PDS). The proposed control law was mathematically solved on a one link and a two-link flexible arm based on the distributed parameter model. Moreover, the authors also presented the Lyapunov function related to the distributed parameter system's total energy. They derived a simple output feedback control law for sensors without an approximated finite-dimensional model at the controller design phase. The convergence and stability proof of the closed-loop DPS were shown using

the invariance principle and the differential operator's characteristic. Experimental results have also been reported to validate the derived model and to show the effectiveness of the proposed control law.

Padhi and Balakrishnan have proposed a variety of control synthesis approach based on neural networks for DPS systems in a series of papers in the early 2000s. The first time it was proposed in (Padhi and Balakrishnan, 2000). The paper's key takeaway is the derivation of the necessary conditions of optimality for DPSs described in discrete domain, followed by the principle of approximate dynamic programming. The infinite time optimal neuro controllers were synthesized based on derived conditions of optimality in the framework of adaptive critic design. The extended version of the work was later published by (Padhi *et al.*, 2001), which includes a comparison with LQR problem for the diffusion equation with a Riccati operator-based solution. For the controller's synthesis, the method proposed two sets of neural networks; the first kind was a set of action networks that captured the mapping between the state and control. Another set of critic networks captures the mapping between the state and co-state.

Another critical part of this survey involves a powerful model reduction technique, known as Proper Orthogonal Decomposition (POD), followed by its usage in numerical approximations techniques such as Galerkin Projection and control design. This method was first introduced by Karhunen (1946) and by Loeve (1945) independently. Hence, this technique is also called Karhunen–Loeve (K–L) expansion. However, the technique gained importance much later. Lumley used POD to study turbulent flows (Lumley, 1981). The method gained popularity when the technique of "snapshots" got incorporated into it by Sirovich (Sirovich, 1987). The power of the technique lies in the fact that it creates a "problem-oriented" basis function, which, when used in conjunction with a reduced basis method (for example, Galerkin's projection), leads to a deficient dimensional representation of the DPS with high accuracy. Several applications based on this technique continue to appear in the literature. A few of them include the paper by (Burns and King, 1998) on the development of a feedback control law of hybrid DPS, the NASA report by (Ravindran, 1999) on the optimal control of fluids, the paper by (Banks and R. C. H., 2000) on the feedback control of a thin shell model and the paper by (Singh and Addington, 2002) on the development of an adaptive feedback

linearizing control for a two-dimensional nonlinear flow.

(Immonen, 2006) designed a controller for linear DPS with bounded control, observation, and feed through operators. Under certain assumptions, this controller achieved asymptotic tracking of arbitrary bounded uniformly continuous reference signals in the presence of disturbances. An independent feedforward feedback controller has been reported where the dynamic feedback part was used to stabilize the closed-loop system consisting of the plant and the controller. In contrast, the feedforward part was tuned using the regulator equations to regulate desired signals.

**Stochastic control**

The design of control algorithms that deliver robust performance in the presence of parametric uncertainty has been an active area of research. Uncertainty in system's parameters is introduced in the systems when the physics that governs the system is known, but the system parameters are either unknown or expected to vary in the operational lifetime. These uncertainties also arise when system models are built on experimental data using system identification techniques, where its transfer function with unknown parameters represents a system plant. As a result of experimental measurements, the values of the transfer function parameters have a range of uncertainty. In either case, the range of parameter variation are assumed to be known which is desired to design controllers that deliver specified performance for these variations. Many articles in the past are found in literature based on controller design considering parametric system uncertainty. Early results include quantitative feedback theory (QFT) (Horowitz, 1982, 1991) and parameter space approach (Siljak, 1989). This was followed by Kharitonov's theorem (Kharitonov, 1978), which gave a simple test to verify the Hurwitz stability of an entire family of interval polynomials. The work of Kharitonov stimulated interested in this area and led to different results that extended well known classical control techniques in the frequency domain (Ackermann, 2002; Bhattacharyya and Keel, 1995; Mansour *et al.*, 2012). In-state space, robust control design methods address parametric uncertainty as structured uncertainty, modelled using linear fractional transformations (LFT) (Doyle, 1982; Safonov, 1982). Many control design techniques have been proposed to design controllers that achieve desired performance with parametric uncer-

tainty. Irrespective of the methods, worst-case analysis is performed in each method to determine the effect of parameter variations on the control system's robustness. Thus, they can be pretty conservative if the parameters rarely assume extreme values. In this way, a control design approach that exploits the distribution of the parameters is expected to be less conservative. This led to a growing interest in combining robust control approaches with stochastic control methods to develop the the field of probabilistic robust control. This field arises from understanding how stochastic system uncertainties affect state trajectories and exploit them to design less conservative robust control algorithms. In the framework of probabilistic robust control, a notion of a risk-adjusted robustness margin trade-off is adopted (Barmish, 2000; Barmish and Shcherbakov, 1996). This framework is more practical as, in many cases, a control system designer may know the distribution of the uncertainty in the system parameters and may be willing to accept a small, well-defined level of risk to obtain larger robustness margins. Moreover, it is possible to stabilize the system for all possible parameter values with a non-zero probability of occurrence and optimize performance concerning the parameters' probability distribution. However, these philosophies for control design result in less conservative controllers. Therefore, it is crucial to consider the range of parameter values and the probability density function of the parameters in the controller design.

Several approaches exist for stability analysis and controller design for stochastic systems (Arnold, 1974; Åström, 2012; Bucy and Joseph, 2005; Chen and Shieh, 1968; Kushner, 1967; Sun, 2006). These results primarily study the evolution of state trajectories when analysis is restricted to the characterization of mean trajectory or deterministic dynamical systems are subjected to stochastic forcing. However, covariance control with the Gaussian excitation has been investigated by Skelton et al. (Skelton and Grigoriadis, 1997). The problem of parametric uncertainty is treated separately from that of stochastic excitation. Much of the work deals with the utilization of sampling-based methods to analyze uncertainty in state trajectories. Barmish et al. use Monte-Carlo based methods to analyze stability and control problems in a Linear Matrix Inequality (LMI) framework (Barmish, 2000), although results are limited to a multidimensional uniform distribution concerning the uncertain parameters. Non-sampling based meth-

ods to determine the evolution of uncertainty in dynamical systems, when there is probabilistic uncertainty in the system parameters were first introduced by Wiener (Wiener, 1938) to model stochastic processes with Gaussian random variables Hermite polynomials were used. It was an extension of Volterra's theory of nonlinear functionals for stochastic systems (Ghanem and Spanos, 2003). According to Cameronand Martin(1947), such an expansion converges in the $L_2$ sense for any arbitrary stochastic process with a finite second moment. This applies to most physical systems. Xiu and Karniadakis (Xiu and Karniadakis, 2002) generalized the result of Cameron–Martin to several continuous and discrete distributions using orthogonal polynomials from the Askey-scheme (Askey and Wilson, 1985) and demonstrated $L_2$ convergence in the corresponding Hilbert functional space. This is popularly known as the generalized polynomial chaos (gPC) framework. The gPC framework has been applied to applications including stochastic fluid dynamics (Hou and Zhou, 2006; Wan and Karniadakis, 2005; Xiu and Karniadakis, 2003) stochastic finite elements and solid mechanics (Ghanem, 1999). Hou et al., in their paper (Hou and Zhou, 2006), showed that gPC based methods are computationally far superior to Monte-Carlo based methods.

The application of gPC for stability analysis was introduced by Hover and Triantafyllou(2006), where the stability of a simple bilinear system was deduced from the gPC expansions. However, their work did not study such stochastic systems in a generalized setting. Furthermore, a systematic framework for controller synthesis for such systems was missing in the article. The gPC framework has also been used in parameter estimation and nonlinear filtering, where better prediction of covariance is achieved using gPC than linear propagation theory, as in extended Kalman filters (Blanchard and Sandu, 2007). In 2007, Prabhakar et al. published a paper where they successfully applied gPC theory to analyze uncertainty propagation in hypersonic flights (Prabhakar *et al.*, 2010) and solved finite-time nonlinear optimal control problems with stochastic parameters (Fisher and Bhattacharya, 2009).

## 2.2 Reduced order models and applications in control theory of beams

It is usually possible to describe physical systems' dynamics by a number of simultaneous linear differential equations with constant coefficients. However, for many processes (like chemical plants and nuclear reactors) or DPSs governed by PDEs, the number of degrees of freedom turns out to be very large. Developing a closed-form solution to these systems is mathematically very challenging and, in some cases, impossible. In such cases, it is common to study these systems by approximating them with a simpler model. Many different methods have been developed to accomplish the purpose by estimating the 'dominant' part of the large system and finding a simpler (or reduced order) system representation with its behaviour akin to the original system. Many methods have been proposed for large scale model order reduction and control using different modelling approaches like model analysis approach, aggregation methods, norm-based methods etc.

A structured approach to model reduction based on model analysis was given by E. J. Davison in (Davison, 1966). The method was based on neglecting the original system's eigenvalues that are farthest from the origin and retain only dominant eigenvalues. These dominant time constants comprise the original system properties in the reduced-order model. The limitation with this method was that it only captured the dynamical system's transient response and failed to capture steady-state response accurately. Noting that Davison's basic model does not give an accurate steady-state response, (Chidambara and Davison, 1967), suggested an approach for model order reduction. In this model, the transient response of the left out states was ignored. Only the steady-state contribution of the states is taken into account to nullify the steady-state error seen in the basic Davison technique. One of the more popular methods for large scale systems order reduction was 'continued fraction' technique. This technique was introduced by Chen and Shieh in (Chen and Shieh, 1968). This methodology was based on the Taylor series expansion of the system's closed-loop transfer function about the origin. The authors used this technique to obtain a reduced-order model for large single-input single-output (SISO) linear time-invariant (LTI) systems based on aggre-

gation methods.

Guyan (Guyan, 1965) introduced the norm-based reduced-order model technique, which is based on model truncation. Parallel to that, (Hurty, 1965) published a paper on dynamic analysis of the structural system using component modes. The component mode synthesis (CMS) method predominantly addresses the problem of analysing the large structures that comprise several small structures (components). CMS solves each component's dynamics separately and uses compatibility conditions at the connection to get response of a large system. In CMS, Degrees of freedom corresponds to physical interference between two components always remain, due to which this method does not give satisfactory reduction.

O'Callahan (O'Callahan, 1989) proposed a modified method named the Improved Reduced System (IRS) method. An additional term is added to the static reduction transformation to make some inertia forces allowance in this approach. Friswell et al. (Friswell and Penny, 1998) extended this idea further to develop another reduction technique, where the system modes were used as the basis for analysis. IRS is an iterative method, which is essentially used to solve non-cyclic systems. O'Callahan et al. (O'Callahan, 1989) published an article in which a novel method based on system mode, similar to IRS, was developed as System Equivalent Reduction Expansion Process (SEREP). In SEREP, the DOFs of the full order systems are divided into master or active and slave or deleted DOFs. The relationship between these two is established using the system's modal properties compared to the equilibrium condition used in the Guyan reduction. Most ROMs reported in the literature can only be applicable to linear systems and not suitable for non-linear systems. Yi dong et al. (Lang *et al.*, 2009). in his paper introduced reduced order modelling technique based on principal component analysis. This is one such technique that holds good for both linear and non-linear systems. Some literature referred to this technique as Proper Orthogonal Decomposition (POD). (Griffith, 2009). One of the important features of the Principal Component Analysis (PCA) based ROM is to develop computationally cost-effective basis sets for dynamical systems (Ma, 1999). PCA has an edge in the structural problems as it can find and rank the dominant coherent modes of the system (Holmes, 1996). Yi–dong et al. (Lang *et al.*, 2009) used data stored from computational fluid dynamics simulations

to develop the ROM based on PCA. This proposed methodology can also be used in signal processing and image processing. PCA decomposes the mixed signals into a set of uncorrelated signals (Stone, 2004).

## 2.3 Gap in research

A brief discussion over the sections of literature reviews on control of beams and on reduced order modelling is presented in the previous section. Based on that discussion, it is observed that a body of work is conducted on control of flexible structures and only a few attension is given on the computational and realtime control aspects. Control needs simulation of a beam finite element model in real time and this can be very time consuming. This thesis tries to fit into this area. An appropriate reduced order modelling is chosen and shown to model well beams in a reduced framework. This thesis, uses SEREP to model beams in a rediced framework and also to extend it for control of vibrations of beams.

## 2.4 Summary

The goal of the present chapter has been to document a thorough study on existing methodologies to develop mathematical model of structural dynamics and vibration control of DPSs. The control algorithms form a broad field of research and development and those related to structural control applications have also been discussed. This chapter also discusses the developments made in reduced order modelling techniques. Finally based on an assessment of the literature, gaps in the field of reduced order model control are identified and reported.

# CHAPTER 3

# Integrated modelling of beams with piezoelectric sensors and actuators

Chapter 2 provided a review to the literature pertaining to control of distributed parameter systems (DPS). One of the important aspect to realize vibration control of DPS is the selection of sensors and actuators. A DPS is better controlled using distributed actuator rather a point actuator. A lot research has been reported in the literature in the area of structural control using smart materials or patches. This chapter starts with an introduction to the smart materials along with historical development of these materials. Moreover development of constitutive equation for piezoelectric materials are also reported. A formulation is developed to integrate the structural dynamics with smart material patch. For simulations, macro fibre composites (MFC) are used as sensors and actuators. Simulation results are reported at the end of the chapter.

## 3.1 Introduction to piezoelectric smart material systems

The study of innovative materials and smart systems has been carried out for decades and is a very diverse discipline. Nevertheless, over the past few decades, the number of materials that promise to be smart based on their interesting material properties has increased significantly (Akhras, 2000). The property that distinguish the smart material from any other materials is that the smart material's energy transfer lies in a multi-physics domain. Thus, smart materials are those that shift energy between multiple physical domains. A physical domain is any physical quantity that can be described by a set of two state variables. Defining physical domains and state variables associated with that domain allows us to define the term coupling more precisely. Coupling between two physical domain exists when a change in the state of one variable in a physical domain causes a change in the state of the variable in different physical domain. Coupling is

usually denoted by a term that combines the names associated with the two physical domains. Some of the materials exhibit a volume change when subjected to an external excitation due to inherent material properties. Such as electro-mechanical and thermo-mechanical properties. This material shrinks, expands, or moves when heated or cooled. Here, a change in the material temperature, a state variable in the thermal domain, can cause a change in the value of strain, a mechanical state variable. The coupling is called thermo-mechanical coupling because the coupling occurs between the thermal and mechanical domains.

The story of the piezoelectric materials traces its way back to the late $19^{th}$ century. The piezoelectric materials got their attention significantly in the early $20^{th}$ century when Paul Langevin, and his co-workers in France built an ultrasonic submarine detector to overcome the submarine threat (Katzir, 2012). This work became the basis of sonar and renowned as one of the most outstanding engineering applications of the piezoelectric effect discovered in the late $19^{th}$ century.

In addition to improvements of sonar devices, developments began to motivate piezoelectric materials as electronic oscillators and filters. The need for more reliable piezoelectric materials having higher coupling co-efficient motivated the development of synthetic piezoelectric materials. Barium titanate $(BaTiO_3)$ was an early synthetic piezoelectric material with both thermal and piezoelectric properties that made this material superior to quartz crystals (Vijatović *et al.*, 2008). Progressions in piezoelectric materials led to their use as ceramic filters for communications devices and other applications, such as cartridges for record players and phonograph.

Work continued developing improved piezoelectric materials in parallel with the crucial developments in electroactive polymers and shape-memory materials (Liu *et al.*, 2009). The discovery of barium titanate $(BaTiO_3)$ motivated researchers to study other material compositions. This work led to the development of lead–zirconate–titanate (PZT) in the late 1950s. PZT exhibits superior piezoelectric properties to barium titanate and remains to be the most widely used piezoelectric material (Yamada *et al.*, 1982; Yang *et al.*, 2000).

### 3.1.1 Application of piezoelectric materials in dynamics and control

By the late 1970, fundamental developments in piezoelectric materials have been made. The last few decades have seen an increasing number of engineering applications being developed that utilize smart materials. One of the most comprehensive commercial applications of piezoelectric materials has been found as motion and force sensors. Piezoelectric crystals are the critical elements for load cells, accelerometers, dynamic pressure sensors, and activators. It is advantageous to use piezoelectric materials for these applications because of their low mass and high mechanical stiffness, which leads to a fast sensing and actuation response. In addition, micro-scale devices, also known as micro-electromechanical systems (MEMs), uses piezoelectric materials as transducing elements. Another most common application of piezoelectric materials has been in microcantilevers (MEMs) for atomic force microscopes (AFMs), which have revolutionized microscopy due to their ability to image and manipulate nanoscale features. A fundamental component of an AFM system is a piezoelectric cantilever beam that vibrates at a very high frequency (in the order of hundreds of kilohertz) and measures interaction forces between the cantilever and the surface or surface topography.

Systems that employ piezoelectric materials have been developed in numerous cases to control noise and vibration. Piezoelectric actuators and sensors have been extensively used in structural vibration control applications because of their excellent sensing and actuating abilities, resulting from their non-intrusive nature and their high electromechanical coupling coefficient. To design the systems for vibration control purposes, piezoelectric transducers are embedded to or bonded in a composite structure. The type of structures that lend themselves to piezoelectric transducers is generally flexible in nature. These system poles are very lightly damped and their transfer functions are of a high order. Control problems associated with these composite systems are by no means trivial. Spillover in control and observations can quickly destabilize a closed-loop system if the controller is only designed for a small number of modes and then implemented on the full-order system (Balas, 1978*b*,*a*). Several techniques have been proposed to address this issue, the most promising of which is based on using collocated

sensors and actuators (Joshi, 1989). The velocity feedback controller has well-known stability properties when implemented on a flexible structure with collocated actuators and sensors (Kumar and Narayanan, 2008*a*; Zorić *et al.*, 2013; Qiu *et al.*, 2009). Although this control is susceptible to spillover, the existence of spillover does not destabilize the closed-loop system. The only difficulty with velocity feedback control is the high controller bandwidth and unnecessary control effort at frequencies far away from the base structure's resonance frequencies. The controller is usually rolled off at higher frequencies by adding extra low pass dynamics. Appropriate care must be taken in adding extra dynamics to the controller since the existence of out bandwidth modes of the structure may result in structural instabilities (Goh and Caughey, 1985). Resonant controllers approximate a velocity feedback controller at and close to the resonance frequencies of a structure. Unlike velocity feedback controllers, their control effort at other frequencies is negligible. However, they are mostly tailored to applications that involve vibration control of a single structural mode and do not extend to multivariable systems. Positive position feedback (PPF) controllers, originally proposed in (Fanson and Caughey, 1990) have been extensively used in structural vibration control applications (Kim *et al.*, 2010; Ferrari and Amabili, 2015; Omidi and Mahmoodi, 2015). These controllers have a structure reminiscent of the system which is being controlled. Compared to resonant controllers, they enjoy a quick roll off outside of the controlled bandwidth, and they are capable of achieving higher levels of performance. Stability conditions for PPF controlled systems, as derived in (Fanson and Caughey, 1990), do not allow for the effect of out of bandwidth modes of the base structure. Consequently, the existence of these modes may destabilize the closed-loop system once the controller is implemented on the structure. Having said this, the natural roll-off of this controller at higher frequencies can often be used to alleviate the effect of out-of-bandwidth modes. Recently it has been demonstrated that the effect of out-of-bandwidth modes of a structure can be captured by adding a feed-through term to the truncated model of the system (Clark, 1997). The corrected model is a close representation of the true system in the controlled bandwidth. The PPF stability condition, as derived in (Fanson and Caughey, 1990), does not allow for the presence of a feed-through term in the structure model. All control techniques introduced thus far are devised for structures with

bonded collocated piezoelectric actuators and sensors. In these systems, a piezoelectric transducer performs a single task of actuation or sensing. These transducers, however, can function simultaneously as actuators and sensors. This property has resulted in the emergence of self-sensing circuits that integrate sensing and actuation functions in the same piezoelectric transducer (Dosch and E. Garcia, 1992; Simmers Jr *et al.*, 2004). If appropriately designed, such circuits can reduce the number of requisite piezoelectric transducers by half.

Applications in noise and vibration control have led to the development of methods for using the same wafer of piezoelectric material simultaneously, as both a sensor and an actuator, thus simplifying the design of vibration control systems (Dosch and E. Garcia, 1992). Piezoelectric and shape memory alloy materials have been used as actuation elements for a deformable aircraft control surface (Jokar *et al.*, 2017). Shape memory alloy actuators for large deflection, low-frequency shape control, while ultrasonic piezoelectric motors were used to control flexible surfaces on the trailing edge of the wing (Sofla *et al.*, 2010).

## 3.2 Piezoelectric material properties

Piezoelectric materials exhibit electro-mechanical coupling, which is indeed helpful for designing systems for sensing and actuation. The coupling is executed when piezoelectric materials undergo an electrical displacement when mechanical stress is applied and produces mechanical strain under an electric field. Since the mechanical-to-electrical coupling was discovered first, energy transfer from mechanical to the electrical domain is termed the direct effect (Leo, 2007). In contrast, the electrical to mechanical coupling is termed the converse piezoelectric effect. Furthermore, piezoelectric materials are also known to exhibit a thermomechanical coupling called the pyroelectric effect (Newnham *et al.*, 1978). This thesis is about vibration control using piezoelectric transducer using their electro-mechanical property. The constitutive development is based on basic mechanisms of electro-mechanical coupling and the pyroelectric effect of piezoelectric material is ignored.

A piezoelectric material's fundamental properties are expressed mathematically as

a relationship between two mechanical variables, stress $(\sigma)$ and strain $(S)$, and two electrical variables, electric displacement $(D)$ and electric field $(E)$. The direct and converse piezoelectric effects are written as the set of linear equations. The strain rate dependent expressions are not considered in this thesis. The expressions for the direct and converse effect can be combined into one matrix expression by expressing the relationship between strain and electric displacement as a function of applied stress and applied field (Leo, 2007).

$$\begin{Bmatrix} S \\ D \end{Bmatrix} = \begin{bmatrix} \mathtt{S} & \mathtt{D} \\ \mathtt{D} & \varepsilon \end{bmatrix} \begin{Bmatrix} \sigma \\ E \end{Bmatrix} \tag{3.1}$$

The top portion of Eq. (3.1) represents the converse piezoelectric effect, while the bottom portion represents the direct effect. The first diagonal term $\mathtt{S}$ represents the mechanical constitutive relationship between strain and stress, whereas the last term on principal diagonal $\varepsilon$ represents the electrical part of the constitutive relation. These constitutive relationships will exist in a material that is either purely dielectric or purely elastic. The off-diagonal terms in matrix Eq. (3.1) represent the electro-mechanical coupling in the material. As the numerical value of the off-diagonal terms increases, it results in such a material that provides more strain for an applied electric field and more electric displacement for applied mechanical stress. For these reasons, the piezoelectric strain coefficient is a critical parameter to compare the relative strength of different types of piezoelectric materials. As the term $\mathtt{D}$ approaches zero, the material exhibits minimal electro-mechanical coupling. Examining Eq. (3.1) it is seen that the coefficient matrix is symmetric. The symmetry of the equation is not simply a coincidence. The symmetry in the coefficient matrix represents reciprocity between the electro-mechanical transduction mechanisms of the material. The symmetry naturally arises while deriving the energy formulation of the piezoelectric constitutive equations.

## 3.3 Constitutive equations for linear piezoelectric material

The fundamental concepts of piezoelectricity have been discussed in the previous section. It can be seen that electromechanical coupling is parameterized by three variables: the piezoelectric strain coefficient D, mechanical compliance S and dielectric permittivity $\varepsilon$. The direct piezoelectric effect and the converse piezoelectric effect are expressed as a relationship between stress $\sigma$, strain $S$, electric field $E$, and electric displacement $D$. The mechanical compliance and dielectric permittivity are functions of the electrical and mechanical boundary condition, respectively.

Consider a cube of piezoelectric material as shown in Fig. 3.1. On each face of the cube three strain components have been defined. One strain component is in the direction perpendicular to the plane of the face, and the other two strain components lie tangent to the plane of the face. The strain component perpendicular to the cube's face is called the normal strain and the strain components tangent to the face are shear components. However, there are no assumptions regarding the direction of the applied electric field or the directions in which the material produces stress or strain in the discussion. Instead, a coordinate system consists of three directions specified numerically, and the standard convention that the three directions are aligned along the poling axis of the material as shown in Fig. 3.1. It is seen from the figure that there are three directions in which an electric field can be applied. These directions are labeled as $E_i$, where $i$ ranges from $1$ to $3$, and are expressed in terms of the electric field vector as,

$$\bar{E} = [E_1 \quad E_2 \quad E_3]^{\mathbf{T}} \tag{3.2}$$

where, $\mathbf{T}$ denotes transpose of vector. Similarly, electric displacement within the material is represented in vector form as,

$$\bar{D} = [D_1 \quad D_2 \quad D_3]^{\mathbf{T}} \tag{3.3}$$

The general relationship between electric field and dielectric displacement variables

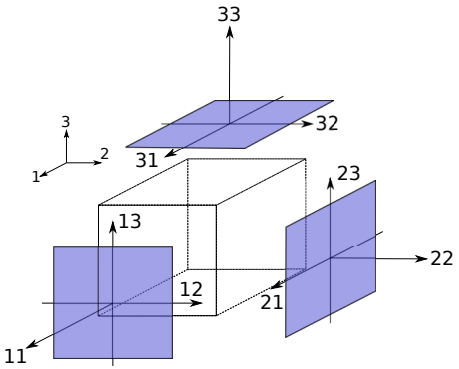



Fig. 3.1: Piezoelectric cube indicating the co-ordinate axes of three dimensional analysis (Leo, 2007).

takes the form as,

$$\begin{aligned} D_1 &= \varepsilon_{11}^{\sigma} E_1 + \varepsilon_{12}^{\sigma} E_2 + \varepsilon_{13}^{\sigma} E_3 \\ D_2 &= \varepsilon_{21}^{\sigma} E_1 + \varepsilon_{22}^{\sigma} E_2 + \varepsilon_{23}^{\sigma} E_3 \\ D_3 &= \varepsilon_{31}^{\sigma} E_1 + \varepsilon_{32}^{\sigma} E_2 + \varepsilon_{33}^{\sigma} E_3 \end{aligned} \tag{3.4}$$

The above expressions can be written in indicial notation as,

$$D_m = \varepsilon_{mn}^{\sigma} E_n \tag{3.5}$$

Where $m$ and $n$ ranges from 1 to 3. The superscript $(.)^{\sigma}$ indicates that the properties are defined at zero stress. The equations related to stress in the $3D$ case can be derived similarly. The general state of stress and strain for the cube of material shows that nine terms are required for complete specification. The strain and stress components are normal to the surfaces of the cube. These are called as principal stress components and are denoted as $\sigma_{11}, \sigma_{22}, \sigma_{33}$ and strain components as $S_{11}, S_{22}, S_{33}$. The in-plane components of stress/strain for each face are called shear components. There are six stress and strain shear components, $\sigma_{12}, \sigma_{13}, \sigma_{23}, \sigma_{21}, \sigma_{32}, \sigma_{31}$ and $S_{12}, S_{13}, S_{23}, S_{21}, S_{32}, S_{31}$ respectively. For a linear elastic material, we can relate the strain to the stress with the

tensor expression,

$$S_{ij} = \mathtt{S}^{E}_{ijkl}\sigma_{kl} \tag{3.6}$$

Where the superscript $(.)^E$ indicates that the properties are defined at zero electrical gradient. The tensor $S_{ijkl}$ represents $81$ mechanical compliance terms. The last step in deriving the three-dimensional constitutive relationships is to specify the coupling between the mechanical and electrical variables. In the most general form, these nine states of strain are related to the three applied electric fields through the expression

$$S_{ij} = \mathtt{D}_{ijn}E_n \tag{3.7}$$

and the three electric displacement terms are related to the mechanical stress through the expression

$$D_m = \mathtt{D}_{mkl}\sigma_{kl} \tag{3.8}$$

Combining these four expressions, we can write the complete set of constitutive equations in strain-charge formulation for a linear piezoelectric material:

$$\begin{aligned} S_{ij} &= \mathtt{S}^{E}_{ijkl}\sigma_{kl} + \mathtt{D}_{ijn}E_n \\ D_m &= \mathtt{D}_{mkl}\sigma_{kl} + \varepsilon^{\sigma}_{mn}E_n \end{aligned} \tag{3.9}$$

The stress-charge formulation is used for numerical analysis of piezoelectric material because it is easy to calculate or measure strains compared to stress.

$$\begin{aligned} \sigma_{ij} &= \mathscr{C}^{E}_{ijkl}\varepsilon_{kl} - e_{kij}E_k \\ D_i &= e_{ikl}\varepsilon_{kl} + \xi^{\epsilon}_{ik}E_k \end{aligned} \tag{3.10}$$

where,

$$\begin{aligned} \mathscr{C}^E &= (\mathtt{S}^E)^{-1} \\ e &= \mathtt{D}(\mathtt{S}^E)^{-1} \\ \xi^\varepsilon &= \xi_0\xi_r - \mathtt{D}(\mathtt{S}^E)^{-1}\mathtt{D}^{\mathbf{T}} \end{aligned} \tag{3.11}$$

Where superscript $(.)^\epsilon$ indicates that the properties are defined at zero strain. The complete set of equations are defined by 81 mechanical compliance constants, 27 piezoelectric strain coefficient and nine dielectric permittivities. In the next section a discussion is made to reduce these equation to a compact form to represent a one dimensional effect.

### 3.3.1 Compact notation for piezoelectric constitutive relations

Eq. (3.10) represents the full set of constitutive relationship for a piezoelectric material. They are written in tensor or indicial notation because they represent a relationship between the nine mechanical field variables (either stress or strain) and the three variables associated with the electric properties.

For analyses of piezoelectric materials based on first principles, it is wise to use the indicial form of the constitutive equations. The indicial form of the constitutive relationships is based on the fact that the stress and strain tensors are symmetric; therefore, With the symmetry of the stress and strain tensor in mind, it is realized that instead of nine independent elements of stress and strain, it can be reduced to only six independent elements. Thus, one can define the following set of new stress and strain components as follow,

$$\begin{aligned} S_1 = S_{11}, \quad S_2 = S_{22}, \quad \sigma_1 = \sigma_{11}, \quad \sigma_2 = \sigma_{22} \\ S_3 = S_{33}, \quad S_4 = S_{23} + S_{32}, \quad \sigma_3 = \sigma_{33}, \quad \sigma_4 = \sigma_{23} = \sigma_{32} \\ S_5 = S_{31} + S_{13}, \quad S_6 = S_{12} + S_{21}, \quad \sigma_5 = \sigma_{31} = \sigma_{13}, \quad \sigma_6 = \sigma_{12} = \sigma_{21} \end{aligned} \tag{3.12}$$

With these definitions, we can write the constitutive equations in a much more compact

notation as shown below,

$$\begin{aligned} S_i &= \mathtt{S}_{ij}^{E} T_j + \mathtt{D}_{ik} E_k \\ D_m &= \mathtt{D}_{mj} T_j + \varepsilon_{mk}^{\sigma} E_k \end{aligned} \tag{3.13}$$

Where $i$ and $j$ range between 1 and 6, and $m$ and $n$ ranges between 1 and 3. The indicial notation highlights that there are only 36 independent elastic constants, 18 piezoelectric strain coefficients, and nine dielectric permittivity values that characterize a piezoelectric material.

## 3.4 Common operating modes of a piezoelectric transducer

Structures embedded with smart materials, e.g. piezoelectric and shape memory alloy, are applied increasingly in several disciplines of technology for vibration and shape control as well as health monitoring applications. Amongst all, piezoelectric materials are the most extensively used in industrial applications due to the number of unique properties these materials possess. Like PZT, Piezoceramics own high structural stiffness enables them to generates strong actuation forces. However, there are several practical limitations for implementing these typically lead-based piezoceramic materials, such as the brittle nature of ceramics, which makes piezoelectric materials susceptible to fracture during handling and bonding procedures, and also their minimal ability to bond around curved surfaces. Another frequently used piezo material is polyvinylidene fluoride (PVDF), which is much more flexible than piezoceramics but possesses limited actuation forces. Integrating favourable properties from both material ideas of a hybrid material consisting of a piezoceramic fibrous phase is embedded in the epoxy matrix phase. The first type of this piezo fibre composite material is referred to as 1–3 composite and is manufactured by Smart Material Corporation (Smart Material GmbH). Another type is active fibre composite (AFC) actuators, which were originally developed by MIT and were the first composite actuators primarily used on structural actuation. The last one, macro-fibre composite (MFC), was developed by NASA Langley Re-

search Center in 1999. The flexible nature of MFC allows the material conforming to a curved surface easily. Additionally, an MFC patch even has larger actuation forces than a PZT patch since the $d_{33}$ effect dominates the actuation mode in MFCs. Because of these beneficial properties of piezoelectric fibre composites, considerable efforts have been made to integrate this material into metal structures for vibration control (Choi and Kim, 2007) and health monitoring (Sodano *et al.*, 2004).

It is advisable to begin with the full constitutive relationships in either indicial notation or compact matrix notation. The constitutive equation (3.9) is first simplified into a smaller subset of expressions for each problem. This allows designers to derive a set of relationships among displacement, force, voltage and charge. These relationships are denoted in the transducer equations for the piezoelectric device.

### 3.4.1 33 - Operating mode (Actuators)

One of the most common operating modes of a piezoelectric device is the direction along the polarization axis. As discussed in the previous section, the convention with piezoelectric materials aligns the three axes of the material in the direction of the polarization vector for the material. For example, consider a small block of piezoelectric material in which the area and thickness are defined as in Fig. 3.2. The $1^{st}$ and $2^{nd}$ directions are in the plane of the transducer. The following assumptions are considered regarding the state of stress and electric field within the material:

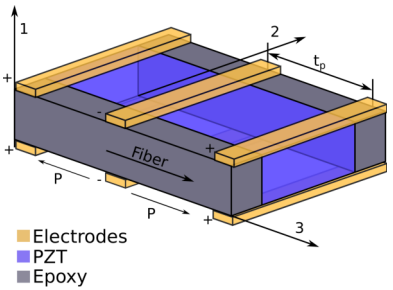



Fig. 3.2: MFC - $d_{33}$ configuration.

$$\sigma_1 = E_1 = 0, \quad \sigma_2 = E_2 = 0, \quad \sigma_4 = 0, \quad \sigma_5 = 0, \quad \sigma_6 = 0 \tag{3.14}$$

whereas only non-zero stress and electric field is in the $3^{rd}$ direction. Under these assumptions, Eq. (3.9) takes form as,

$$\begin{aligned} S_1 &= -\frac{\nu_{13}}{Y_1^E}\sigma_3 + \mathtt{d}_{13}E_3, \quad S_2 = -\frac{\nu_{23}}{Y_1^E}\sigma_3 + \mathtt{d}_{23}E_3 \\ S_3 &= \frac{1}{Y_1^E}\sigma_3 + \mathtt{d}_{33}E_3, \quad D_3 = \mathtt{d}_{33}\sigma_3 + \varepsilon_{33}^{\sigma}E_3 \end{aligned} \tag{3.15}$$

Equation (3.15) define the state of strain and electric displacement in the piezoelectric material as a function of the stress and electric field prescribed. In this form of the constitutive equations, the stress and electric field are the independent variables, and strain and electric displacement are the dependent variables.

For piezoelectric actuator, a $d_{33}$ piezoelectric device, the state of the piezoelectric material is analysed in the $3^{rd}$ direction. In that case, mechanical and electrical boundary conditions are prescribed by specifying only $S_3$ and $D_3$ from Eq. (3.15) and solve for the stress and electric field.

**Transducer equations for a 33-piezoelectric device**

Developing a simple mathematical formulation while designing a device that uses a $d_{33}$-mode piezoelectric transducer, constitutive relationships are further reduced to a set of relationships that relate directly to the force $\mathfrak{f}$, displacement $u$, voltage $\nu$, and charge $q$. Assuming that the strain in the three directions is uniform,

$$S_3 = \frac{u_3}{t_p} \tag{3.16}$$

where $u_3$ is strain in $3^{rd}$ direction and $t_p$ is thickness of piezoelectric material. If the stress is uniform over the surface, then

$$\sigma_3 = \frac{\mathfrak{f}}{A_p} \tag{3.17}$$

where $\mathfrak{f}$ is normal force in $N$ applied over a surface with an area of $A_p$. A electric field assumed to be applied uniformly is related to the applied potential ($\nu$) through the

expression

$$E_3 = \frac{\nu}{t_p} \tag{3.18}$$

Assuming a no wear and tear of electrode surface, the electric displacement can be assumed to be uniform and related to the charge as follow,

$$D_3 = \frac{q}{A_p} \tag{3.19}$$

Substituting expressions Eq. (3.16) to Eq. (3.19) into Eq. (3.15) yields

$$\frac{u_3}{t_p} = \frac{1}{Y_3^E}\frac{\mathfrak{f}}{A_p} + \mathfrak{d}_{33}\frac{\nu}{t_p} \tag{3.20}$$

$$\frac{q}{A_p} = \mathfrak{d}_{33}\frac{\mathfrak{f}}{A_p} + \varepsilon_{33}^{\sigma}\frac{\nu}{t_p} \tag{3.21}$$

Rearranging the above expressions fetches the transducer equations for a $d_{33}$-mode piezoelectric device:

$$u_3 = \frac{t_p}{Y_3^E A_p}\mathfrak{f} + \mathfrak{d}_{33}\nu \tag{3.22}$$

$$q = \mathfrak{d}_{33}\mathfrak{f} + \frac{\varepsilon_{33}^{\sigma} A_p}{t_p}\nu \tag{3.23}$$

Important design information can be derived from Eq. (3.22). The free displacement, $\delta_o$, of the transducer is equal to

$$\delta_o = u_3|_{\mathfrak{f}=\mathfrak{o}} = \mathfrak{d}_{33}\nu \tag{3.24}$$

and is the displacement of the material in the $3^{rd}$ direction when there is no restraining force. Similarly, the blocked force is

$$\mathfrak{f}_{b1} = -\mathfrak{f}|_{u_3=0} = \mathfrak{d}_{33} Y_3^E \frac{A_p}{t_p}\nu \tag{3.25}$$

The development of transducer equations for sensor is discussed in next section.

### 3.4.2 31 - Operating mode (Sensors)

Another common use of a piezoelectric device is to apply an electric field in the $3^{rd}$ direction but utilize the stress and strain produced in the $1^{st}$ direction to create extension or bending in the material. The most common devices that employs this methodology is piezoelectric sensors. In the use of the $d_{31}$ operating mode of a piezoelectric following assumptions are made,

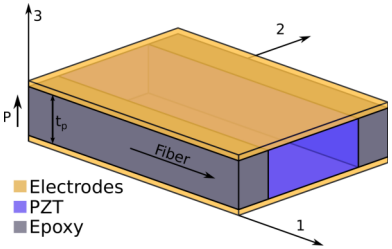



Fig. 3.3: MFC - $d_{31}$ configuration

$$E_1 = 0, \quad \sigma_3 = 0, \quad \sigma_2 = E_2 = 0, \quad \sigma_4 = 0, \quad \sigma_5 = 0, \quad \sigma_6 = 0 \tag{3.26}$$

Under these assumptions, the constitutive equations are reduced to

$$\begin{aligned} S_1 &= -\frac{1}{Y_1^E}\sigma_1 + \mathtt{d}_{13}E_3, \quad S_2 = -\frac{\nu_{21}}{Y_1^E}\sigma_1 + \mathtt{d}_{23}E_3 \\ S_3 &= -\frac{\nu_{31}}{Y_1^E}\sigma_1 + \mathtt{d}_{33}E_3, \quad D_3 = \mathtt{d}_{31}\sigma_1 + \varepsilon_{33}^{\sigma}E_3 \end{aligned} \tag{3.27}$$

where the piezoelectric strain coefficients are symmetric. As is the case with a $d_{33}$ transducer, the reduced constitutive equations (3.27) for a $d_{31}$ transducer have two independent variables and four dependent variables. In general, the equation (3.27) can further be reduced to two equations as the nature of the problem.

$$\begin{aligned} S_1 &= \frac{1}{Y_1^E}\sigma_1 + \mathtt{d}_{13}E_3 \\ D_3 &= \mathtt{d}_{13}\sigma_1 + \varepsilon_{33}^{\sigma}E_3 \end{aligned} \tag{3.28}$$

The transducer relations for a single piezoelectric element (i.e., sensor) operating in the $d_{31}$ mode can be derived by substituting the relationships

$$S_1 = \frac{u_1}{L_p}, \qquad \sigma_1 = \frac{\mathfrak{f}}{w_p t_p} \tag{3.29}$$

$$E_3 = \frac{\nu}{t_p} \qquad D_3 = \frac{q}{w_p L_p} \tag{3.30}$$

into Eq. (3.28):

$$\begin{aligned} \frac{u_1}{L_p} &= \frac{\mathfrak{f}}{Y_1^E w_p t_p} + \mathtt{d}_{13}\frac{\nu}{t_p} \\ \frac{q}{w_p L_p} &= \mathtt{d}_{31}\frac{\mathfrak{f}}{w_p t_p} + \varepsilon_{33}^{\sigma}\frac{\nu}{t_p} \end{aligned} \tag{3.31}$$

Rearranging the terms yields

$$\begin{aligned} u_1 &= \frac{L_p}{Y_1^E w_p t_p}\mathfrak{f} + \frac{\mathtt{d}_{13} L_p}{t_p}\nu \\ q &= \frac{\mathtt{d}_{31} L_p}{t_p}\mathfrak{f} + \frac{\varepsilon_{33}^{\sigma} w_p L_p}{t_p}\nu \end{aligned} \tag{3.32}$$

where $w_p$ and $L_p$ is the width and length of electrode respectively. From these expressions the free displacement and blocked force can be derived same as previous section. The formulation for same can be given as,

$$\delta_o = \frac{\mathtt{d}_{13} L_p}{t_p}\nu \tag{3.33}$$

$$\mathfrak{f}_{bl} = \mathtt{d}_{13} Y_1^E w_p \nu \tag{3.34}$$

Comparing Eq. (3.24) and Eq. (3.33), it is seen that the free displacement of a 31-mode transducer can be scaled by increasing the length-to-thickness ratio, $L_p/t_p$ , whereas the free displacement in the $d_{33}$-mode is only a function of the piezoelectric strain coefficient.

## 3.5 Numerical modelling of beam integrated with sensors and actuators

Mathematical modelling of piezoelectric patches has been derived in previous section. The integration of these patches to the linear flexible cantilever beam is developed in this section. Moreover the developed formulation is validated using simulation software.

### 3.5.1 Modelling of an Euler-Bernoulli beam

The formulation for a homogeneous, isotropic, thin Euler-Bernoulli (EB) beam with multiple collocated piezoelectric actuator and sensor pairs as shown in Fig. 3.4 is discussed here. The piezoelectric layers are perfectly bonded with the beam using adhesive. The adhesive layer being very thin compared to other structure dimensions is neglected in developing the mathematical formulation. However, the contribution of piezoelectric actuators and sensors to the beam's mass and stiffness matrices has been considered in the formulation. The linear beam is modelled using an EB beam theory. The governing equation for transverse vibration of beam without piezoelectric actuator

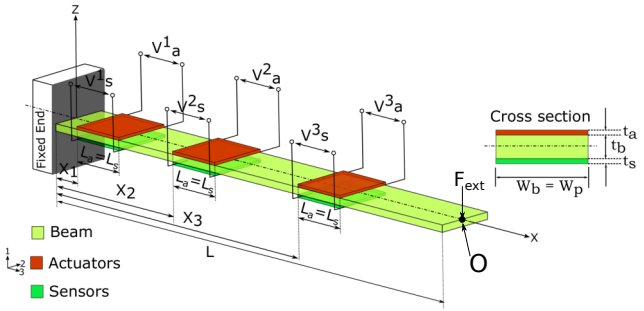



Fig. 3.4: Schematic of the cantilever beam with sensors and actuators in collocated configuration.

and sensor is given (Rao, 2007) by,

$$\frac{\partial^2 M}{\partial x^2} + \rho A \frac{\partial^2 w}{\partial t^2} = q(x,t) \tag{3.35}$$

where,

$$M = EI\frac{\partial^2 w}{\partial x^2} \tag{3.36}$$

Where $M$ denotes the flexural moment across the cross-section of the beam, $w$ is deflection of beam in $Z$ direction, $\rho$, $A$ and $q$ denote the density of beam material, cross-sectional area of the beam and transverse force, respectively. The integration of piezoelectric properties with that of governing partial differential Eq. (3.35) is discussed in the next section.

### 3.5.2 Piezoelectric constitutive relations

In this study, the linear piezoelectric constitutive relation Eq. (3.10) is used to incorporate sensors and actuators properties with the beam. The constitutive relation for piezoelectric material is considered from piezoelectric stress-charge formulation and is given as,

$$\begin{aligned} \sigma_{ij} &= \mathscr{C}^{E}_{ijkl}\varepsilon_{kl} - e_{kij}E_k \\ D_i &= e_{ikl}\varepsilon_{kl} + \xi^{\epsilon}_{ik}E_k \end{aligned} \tag{3.37}$$

where, $i, j, k$ and $l$ ranges between 1 and 3. The superscript $(.)^E$ and $(.)^\epsilon$ denotes that the properties are measured at the zero electric gradient and zero strain. $\sigma$ is the stress tensor generated due to the strain $\varepsilon$ and electric field $E$. $D$ is the electric displacement, $e$ is the piezoelectric tensor, and $\xi$ is the dielectric tensor. The electric field can be defined as a gradient of electric potential $\Phi$. For low electric field, linear constitutive relations given in Eq. (3.37) are applicable. It should be noted that the hysteresis and creep effects of the piezoelectric material are not considered in this study. The electric field components are related to the electrostatic potential $\varphi$ by the equation,

$$E_3 = -\varphi_{,3} = -\frac{\partial \varphi}{\partial z} \tag{3.38}$$

Based on Eq. (3.37), the stress-strain relation for one dimensional structure with uniaxial loading can be derived as (Kumar and Narayanan, 2008*b*),

$$\sigma_{11} = c_{11}\varepsilon_{11} - e_{31}E_3 \tag{3.39}$$

Where $c_{11}$ is the Young's modulus of piezoelectric material, $\phi_a$ is the voltage applied in the piezoelectric material, and $t_a$ is the thickness of the piezoelectric material. The governing differential Eq. (3.35) for a beam with the addition of the actuators and sensors becomes,

$$\frac{\partial^2 \left(M + \sum_{i=1}^{m} M_p^i \mathbb{H}_i(x)\right)}{\partial x^2} + \left(\rho A + \sum_{i=1}^{m} \rho_p^i A_p^i \mathbb{H}_i(x)\right) \frac{\partial^2 w}{\partial t^2} = q(x,t) + U(x,t) \tag{3.40}$$

Where $U$ is the force due to the piezoelectric actuator, the subscript $(\cdot)_p^i$ refers to the properties of $i^{th}$ piezoelectric material. The actuation part of the piezoelectric equations is separated and combined in the control force. Eq. (3.40) is governing differential equation fourth order in space and second order in time. This is also termed as strong from of differential equation. Eq. (3.35) includes the dynamics of linear cantilever structures while as contribution of piezoelectric transducers to the dynamics of system is captured in Eq. (3.40). The control force $U$ is defined as,

$$U(t) = \sum_{i=1}^{m} r_i \, \bar{U}_i(\phi_{ai}) \, \mathbb{H}_i(x) \tag{3.41}$$

$$\bar{U}_i = \frac{\partial^2}{\partial x^2} \left(z e_{31} (-\varphi_{,3})_i\right) \tag{3.42}$$

$$\mathbb{H}_i(x) = \begin{cases} 1 & x \in A_i \\ 0 & otherwise \end{cases} \tag{3.43}$$

where, $r_i$ is the weight of $i^{th}$ actuator, $\bar{U}_i$, $A_i$ and $m$ are the actuation force, domain of the $i^{th}$ actuator and number of collocated sensor/actuator pair respectively.

## 3.6 FE modelling for integrated beam-actuator-sensing system

The weak form of the equation of motion for a beam with piezoelectric transducers is derived by using Hamilton's Principle as follow,

$$\delta \int_{t1}^{t2} (T - \Pi + W_{nc}) dt = 0 \tag{3.44}$$

Where the total kinetic energy $(T)$ of structure is given by

$$T = \int_{v} \frac{1}{2} \rho (\dot{u}^2 + \dot{w}^2) dv \tag{3.45}$$

The variation in kinetic energy of system is given by

$$\delta T = \int_{v} \rho (\delta u \ddot{u} + \delta w \ddot{w}) dv \tag{3.46}$$

The energy stored in the PZT is due to internal stresses, and the electrostatic field developed as a result of external loads. Thus the total internal energy $\Pi$ is the combination of elastic and electrostatic energy. The energy stored in a body as it is being deformed under the action of an external load is the elastic energy $\pi_{ie}$ of deformation and is given by,

$$\pi_{ie} = \int_{v} \frac{1}{2} \sigma_{ij} \epsilon_{kl} dv \tag{3.47}$$

The energy stored due to work done by an external force in moving a charge is given by,

$$\pi_{E} = \int_{v} \frac{1}{2} E_k D_i dv \tag{3.48}$$

Total potential energy $(\Pi)$ of the electro-elastic system is a sum of internal strain energy $(\pi_{ie})$ and the electric strain energy $(\pi_E)$ and is obtained using piezoelectric constitutive

relations,

$$\begin{aligned}\Pi &= \int_v \frac{1}{2}\sigma_{ij}\epsilon_{kl}dv - \int_v \frac{1}{2}E_k D_i dv \\ &= \int_v \left[\frac{1}{2}\mathscr{C}_{ijkl}\epsilon_{ij}\epsilon_{kl} + e_{kij}E_i\epsilon_{jk} - \frac{1}{2}\xi^{\varepsilon}_{ij}E_iE_j\right]dv\end{aligned} \tag{3.49}$$

The work done due to non-conservative forces is a sum of the work done by external transverse force ($f_t$), the axial force ($f_a$) and electric potential ($\varphi$) which are given as,

$$W_{nc} = \int_s f_t w ds + \int_s f_a u ds - \int_s \sigma\phi ds \tag{3.50}$$

The variation in non-conservative forces is given by,

$$\delta W_{nc} = \int_s f_t \delta w ds + \int_s f_a \delta u ds - \int_s \sigma\phi\delta\phi ds \tag{3.51}$$

### 3.6.1 Finite element formulation

A two noded beam element with three mechanical variables $(u, w, \theta)$ and one electric degree of freedom ($\varphi$) at each node is used. The axial displacement $u$ is expressed in the nodal displacement in the finite element model as follows,

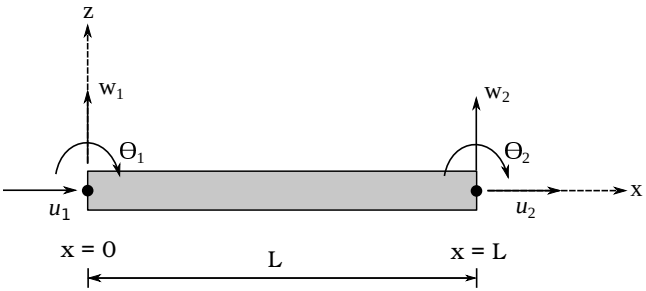



Fig. 3.5: A finite beam element with corresponding DoFs.

$$u(x) = \sum_{j=1}^{2} u_j \psi_j(x) = [\psi_1 \quad \psi_2] \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} \tag{3.52}$$

where the linear interpolation functions given by,

$$\psi_1 = 1 - \frac{x}{L}, \quad \psi_2 = \frac{x}{L} \tag{3.53}$$

The transverse displacement $w$ and the rotation $\theta$ are expressed in the finite element model by a Hermite cubic interpolation shape functions such as,

$$w(x) = \sum_{j=1}^{4} w_j \phi_j(x) = [\phi_1 \quad \phi_2 \quad \phi_3 \quad \phi_4] \begin{bmatrix} w_1 \\ \theta_1 \\ w_2 \\ \theta_2 \end{bmatrix} \tag{3.54}$$

where,

$$\begin{aligned} \phi_1 &= 1 - \frac{3x^2}{L^2} + \frac{2x^3}{L^3} \quad & \phi_2 &= x - \frac{2x^2}{L} + \frac{x^3}{L^2} \\ \phi_3 &= \frac{3x^2}{L^2} - \frac{2x^3}{L^3} & \phi_4 &= -\frac{x^2}{L} + \frac{x^3}{L^2} \end{aligned} \tag{3.55}$$

For the piezoelectric element, the electric field is treated as the electric degree of freedom like a generalized displacements degrees of freedoms (Agrawal *et al.*, 1997).

$$\varphi(z) = \sum_{j=1}^{2} \varphi_j \vartheta_j(z) = [\vartheta_1 \quad \vartheta_1] \begin{bmatrix} \varphi_1 \\ \varphi_2 \end{bmatrix} \tag{3.56}$$

where,

$$\vartheta_1 = \frac{1}{2} + \frac{z}{h} \qquad \vartheta_2 = \frac{1}{2} - \frac{z}{h} \tag{3.57}$$

The electric potential is considered as a function of the thickness and the length of the element (Benjeddou *et al.*, 2000), thus by the product of Eq. (3.53) and Eq. (3.57) and impose the homogenous boundary condition on the bottom surface to eliminate the rigid

body modes. The shape functions finally take the form,

$$\vartheta_1 = \left(\frac{1}{2} + \frac{z}{h}\right)\left(1 - \frac{x}{L}\right), \quad \vartheta_2 = \left(\frac{1}{2} - \frac{z}{h}\right)\left(\frac{x}{L}\right) \tag{3.58}$$

## 3.7 Development of equations of motion

To minimize the total strain energy $U$, the first variation is taken with respect to the axial and transverse displacements and the electric potential. Thus, the first variation of total potential energy Eq. (3.49) is represented by,

$$\delta U = \int_v \left[\delta\epsilon_{ij}\mathscr{C}_{ijkl}\epsilon_{kl} + \delta\epsilon_{jk}e_{kij}E_i + \delta E_i e_{kij}\epsilon_{ij} - \delta E_i \xi_{ij} E_j\right] dv \tag{3.59}$$

By substituting the linear strain relation of EB beam into Eq. (3.59) the total variation takes from as,

$$\begin{aligned}
\delta U = &\int_v \mathscr{C}_{ijkl}\left(\frac{\partial u}{\partial x}\right)\left(\delta\frac{\partial u}{\partial x}\right) dv - \int_v \mathscr{C}_{ijkl}\left(\frac{\partial u}{\partial x}\right)\left(\delta z\frac{\partial^2 w}{\partial x^2}\right) dv - \\
&\int_v \mathscr{C}_{ijkl}\left(z\frac{\partial^2 w}{\partial x^2}\right)\left(\delta\frac{\partial u}{\partial x}\right) dv - \int_v \mathscr{C}_{ijkl}\left(z\frac{\partial^2 w}{\partial x^2}\right)\left(\delta z\frac{\partial^2 w}{\partial x^2}\right) dv + \\
&\int_v e_{kij}\left(\frac{\partial \varphi}{\partial z}\right)\left(\delta z\frac{\partial u}{\partial x}\right) dv + \int_v e_{kij} z\left(\frac{\partial \varphi}{\partial z}\right)\left(\delta z\frac{\partial^2 w}{\partial x^2}\right) dv + \\
&\int_v \epsilon_{ij}\left(\frac{\partial \varphi}{\partial z}\right)\left(\delta\frac{\partial \varphi}{\partial z}\right) dv
\end{aligned} \tag{3.60}$$

By substituting the shape functions, Eq. (3.52), Eq. (3.54), and Eq. (3.56) into Eq. (3.60) the first variation of the strain energy and the virtual work expressions take the forms,

$$\delta U = \int_v \left[ \mathscr{C}_{ijkl} \left( \frac{\partial \psi_j}{\partial x} \frac{\partial \psi_i}{\partial x} \right) u_i - \mathscr{C}_{ijkl} z \left( \frac{\partial \psi_j}{\partial x} \frac{\partial^2 \phi_i}{\partial x^2} \right) w_i + e_{kij} \left( \frac{\partial \psi_j}{\partial x} \frac{\partial \vartheta_i}{\partial x} \right) \varphi_i \right] \delta u_i dv -$$
$$\int_v \left[ \mathscr{C}_{ijkl} z \left( \frac{\partial^2 \phi_j}{\partial x^2} \frac{\partial \psi_i}{\partial x} \right) u_i - \mathscr{C}_{ijkl} z^2 \left( \frac{\partial^2 \phi_j}{\partial^2 x} \frac{\partial^2 \phi_i}{\partial x^2} \right) w_i + e_{kij} z \left( \frac{\partial^2 \phi_j}{\partial x^2} \frac{\partial \vartheta_i}{\partial z} \right) \varphi_i \right] \delta w_i dv +$$
$$\int_v \left[ e_{kij} \left( \frac{\partial \vartheta_j}{\partial z} \frac{\partial \psi_i}{\partial x} \right) u_i - e_{kij} z \left( \frac{\partial \vartheta_j}{\partial z} \frac{\partial^2 \phi_i}{\partial x^2} \right) w_i + \xi_{ij} \left( \frac{\partial \vartheta_j}{\partial z} \frac{\partial \vartheta_i}{\partial z} \right) \varphi_i \right] \delta \varphi_i dv \tag{3.61}$$

By substituting the shape functions into total kinetic energy Eq. (3.45) it takes following from,

$$T = \frac{1}{2} \left[ \dot{u}_1 \, \dot{u}_2 \, \dot{w}_1 \, \dot{\theta}_1 \, \dot{w}_2 \, \dot{\theta}_2 \right] \left( \rho A \int_0^L \begin{bmatrix} \psi_1 \\ \psi_2 \\ \phi_1 \\ \phi_2 \\ \phi_3 \\ \phi_4 \end{bmatrix} \left[ \psi_1 \, \psi_2 \, \phi_1 \, \phi_2 \, \phi_3 \, \phi_4 \right] dx \right) \begin{bmatrix} \dot{u}_1 \\ \dot{u}_2 \\ \dot{w}_1 \\ \dot{\theta}_1 \\ \dot{w}_2 \\ \dot{\theta}_2 \end{bmatrix} \tag{3.62}$$

After substituting shape functions, Eq. (3.52), Eq. (3.54), and Eq. (3.56) into Eq. (3.51) the first variation of the work done due to non conservative force takes form as,

$$\delta W_{nc} = \iint_R \left[ \psi_i f_a \right] \delta u_j dy dz - \iint_R \left[ \vartheta_j \sigma \right] \delta \varphi_i dx dy + \iint_R \left[ \phi_j f_t - z \frac{\partial \phi_j}{\partial x} t \right] \delta w_i dx dy \tag{3.63}$$

After performing integration through thickness and taking all variation to zero, the elements of the stiffness matrix are represented by,

$$K_{11} = \int_{x_a}^{x_b} A_{xx} \left( \frac{\partial \psi_j}{\partial x} \frac{\partial \psi_i}{\partial x} \right) dx \qquad K_{12} = \int_{x_a}^{x_b} B_{xx} \left( \frac{\partial \psi_j}{\partial x} \frac{\partial^2 \phi_i}{\partial x^2} \right) dx$$

$$K_{13} = \iint_R \int e_{13} \left( \frac{\partial \psi_j}{\partial x} \frac{\partial \vartheta_i}{\partial z} \right) dxdydz \qquad K_{21} = \int_{x_a}^{x_b} B_{xx} \left( \frac{\partial^2 \phi_j}{\partial x^2} \frac{\partial \psi_i}{\partial x} \right) dx$$

$$K_{22} = \int_{x_a}^{x_b} D_{xx} \left( \frac{\partial^2 \phi_j}{\partial^2 x} \frac{\partial^2 \phi_i}{\partial x^2} \right) dx \qquad K_{23} = \iint_R \int e_{13} z \left( \frac{\partial^2 \phi_j}{\partial x^2} \frac{\partial \vartheta_i}{\partial z} \right) dxdydz$$

$$K_{31} = \iint_R \int e_{13} \left( \frac{\partial \vartheta_j}{\partial z} \frac{\partial \psi_i}{\partial x} \right) dxdydz \qquad K_{32} = \iint_R \int e_{13} z \left( \frac{\partial \vartheta_j}{\partial z} \frac{\partial^2 \phi_i}{\partial x^2} \right) dxdydz$$

$$K_{33} = \iint_R \int \xi_{33} \left( \frac{\partial \vartheta_j}{\partial z} \frac{\partial \vartheta_i}{\partial z} \right) dxdydz \tag{3.64}$$

where, $A_{xx}$, $B_{xx}$ and $D_{xx}$ are the extensional, extensional-bending and bending stiffness of beam element,

$$(A^e_{xx}, B^e_{xx}, D^e_{xx}) = \int_{A_e} E^e \left(1, z, z^2\right) dA \tag{3.65}$$

The equation of motion of the whole structure system is represented by,

$$\begin{bmatrix} M_{ww} & 0 \\ 0 & 0 \end{bmatrix} \begin{Bmatrix} \ddot{W} \\ \ddot{\varphi} \end{Bmatrix} + \begin{bmatrix} [K_{ww}] & [K_{w\varphi}] \\ [K_{\varphi w}] & [K_{\varphi\varphi}] \end{bmatrix} \begin{Bmatrix} W \\ \varphi \end{Bmatrix} = \begin{Bmatrix} F \\ Q \end{Bmatrix} \tag{3.66}$$

where,

$$K_{ww} = \begin{bmatrix} K_{11} & K_{12} \\ K_{21} & K_{22} \end{bmatrix};\ K_{w\varphi} = \begin{bmatrix} K_{13} \\ K_{23} \end{bmatrix};\ K_{\varphi w} = \begin{bmatrix} K_{31} & K_{32} \end{bmatrix};\ K_{\varphi\varphi} = [K_{33}] \tag{3.67}$$

Where, $M_{ww}$ is the global mass matrix, $K_{ww}$ is the global stiffness matrix corresponding to mechanical degrees of freedom, $K_{w\varphi}$ is the global stiffness matrix due to electromechanical coupling and $K_{\varphi\varphi}$ the global stiffness matrix due to electrical dofs alone. Also, $W$ is the vector that consists of transverse displacement and rotation of the struc-

tural part of the system, and $\varphi$ is a vector of electrical dofs corresponding to piezoelectric actuators. Here, $F$ is the externally applied load on the system, and $Q$ is the vector of electrical charge.

## 3.8 Numerical and simulation result

The governing equation of motion of an EB beam integrated with actuators and sensors are developed in subsection 3.5.2 . The exact solution to Eq. (3.40) is very challenging.

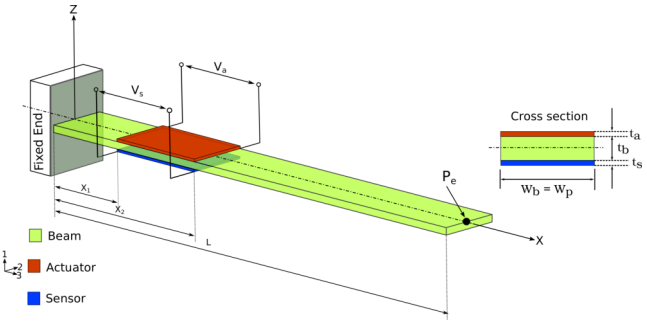



Fig. 3.6: Flexible cantilever beam with piezoelectric transducer

The most common ways to solve this dynamical system is to discretize it using FE approach as discussed in previous section. The discretization leads to set of second order differential equations. The Eq. (3.66) is solved simultaneously in order to get dynamics of system. This approach is coded in numerical software (MATLAB) and to validate the code the same system is solved in commercial FE software (ABAQUS). An example of flexible cantilever beam with piezoelectric actuator and sensor pair, as shown in Fig. 3.6 has been taken to validate the efficacy of the developed code. The beam's geometric dimensions has been taken to be $400mm \times 10mm \times 0.3mm$ and other mechanical and piezoelectric properties are taken from table 4.1. A two-node beam element with three mechanical and one electric degree of freedom at each node is used to develop MATLAB code.

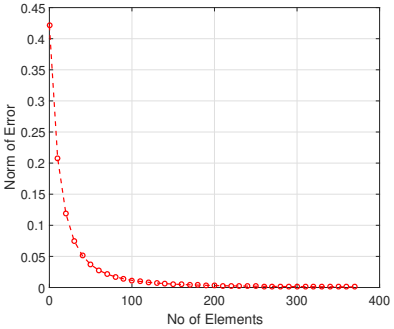



Fig. 3.7: Convergence of number of elements for finite element analysis.

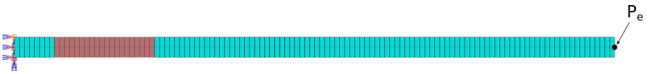



Fig. 3.8: Flexible cantilever beam with piezoelectric transducer - Mesh

Fig. 3.7 shows the convergence study of the number of elements in the finite discretization of the system. It can be seen that after 200 elements, the error norm of natural frequencies has become constant. Hence, based on the convergence study, the system has been discretized using 200 elements in the current analysis. To maintain compatibility conditions in FE analysis, the beam is discretized in such a way that number of elements in beam, where there are piezoelectric patches, are same as those of elements in patches. The same two noded beam element cannot be used to model the system in

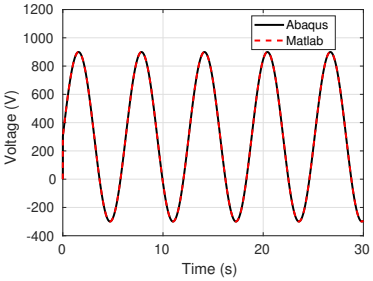
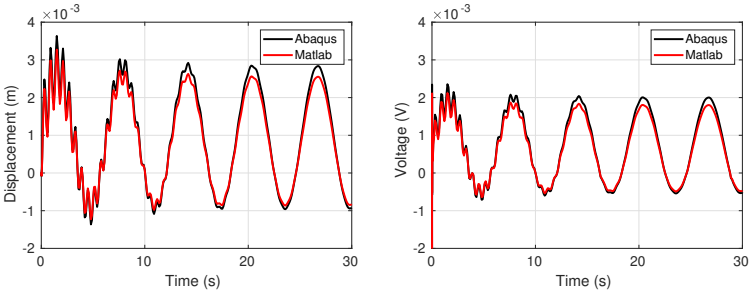



Fig. 3.9: Voltage applied to the actuator




(a) Transverse displacement (at the free end) obtained through ABAQUS and MATLAB

(b) Comparison of sensor output obtained through ABAQUS with that of MATLAB

Fig. 3.10: Comparison of output obtained through Numerical and Simulation software

ABAQUS because of unavailability of two noded piezoelectric element in the software. Therefore a 4-node thin linear shell (S4R) is used to model the base structure as thickness of the system considered under study is very small. An 8-node linear piezoelectric brick element (C3D8E) is used to model sensor and actuator in ABAQUS. The shell element of base structure and piezoelectric transducers are tied using 'TIE' constraints to maintain compatibility conditions. The system is made cantilever by arresting all its DoFs from one side as shown in Fig. 3.8. The developed MATLAB code is validated using the model developed in simulation software ABAQUS. Here, the piezoelectric actuator is used as an external actuating source. The actuator is subjected to periodic electric loading as shown in Fig. 3.9 in the both cases. The dynamical response is measured at the free end node $P_e$. It can be noticed from Fig. 3.10a that the dynamic response obtained through simulation software shows very good agreement with that obtained from in-house developed code. Moreover from Fig. 3.10b it is clear that the voltage generated from sensor in both the cases shows a very good match.

## 3.9 Summary

The brief history of piezoelectricity is first discussed followed by it's applications and material properties. The electro-mechanical constitutive relationship along with common operating modes of piezo materials are also documented. This chapter also provides the numerical technique to incorporate piezoelectric material with linear beams.

The developed numerical model is compared with the commercial software and the results of this validation study are reported in the end of the chapter.

# CHAPTER 4

# Reduced order model integrated control of cantilever beams

The electro-mechanical modelling of a linear cantilever beam is discussed in Chapter 3. This finite element models are computationally expensive and doing real time analysis of control of these systems becomes challenging. In this chapter the finite element model is further reduced using reduced order modelling technique. Although there are many ROM techniques available SEREP is chosen because of it's powerful model order reduction. The control algorithm is developed using this reduced order model and this algorithm is tested on flexible cantilever beam with piezoelectric patches. The chapter ends with findings of the proposed algorithm.

## 4.1 Introduction

Flexible structures are widely used in many applications ranging from robotics, biomechanics, wind farms, aerial vehicle to space structure applications. These structures have relatively less structural/material damping that delays the decay of amplitude of vibrations (Padhi *et al.*, 2001). This may hamper the proper functioning and/or efficiency of the complete system (Ali and Ramaswamy, 2009). To overcome this problem, active, passive or semi-active mechanisms are deployed for structural vibration control.

The design space application in MEMs, soft robotics, flexible aircraft wings demanded advanced features that can only be catered using a real-time control of their flexible parts (Chen and Tan, 2010; Sodano *et al.*, 2004). Distributed control is achieved using smart materials like piezocomposites (Sodano *et al.*, 2004), shape-memory actuators (Baz and McCoy, 1990), shape-memory polymers (Saadat *et al.*, 2002). For example, structures embedded with piezoelectric sensors and actuators are used for active vibration control of flexible structure (Julai and Tokhi, 2012; Dosch and Inman, 1995; Hu and Ng, 2005).

These flexible structures are mathematically modelled as a continuous system using partial differential equations. To obtain a closed-form solution is a tedious phenomenon. Numerical techniques are employed to get an approximate solution. There are many numerical techniques available (Ames, 2014) among them Finite Element (FE) method is used widely in problems related to structural dynamics. To estimate the solution numerically using the FE approach, the strong form of the equations is written into a weak form. This full-order FE model contains a large number of degrees of freedom (dofs), and its solution becomes computationally exhaustive and time-consuming for real-time applications. To bypass the complexity associated with computations without sacrificing accuracy, reduced-order models (ROMs) are used for analysis. A wide range of methods has been reported in the literature for linear as well as non-linear dynamical systems. For example, modal truncation (Hughes and Skelton, 1981), balanced truncation (Willcox and Peraire, 2002) are used for linear systems, and proper orthogonal decomposition (POD), which is also known as principal component analysis (PCA) (Hsieh, 2001; Kerschen and Bergman, 2005) is used for non-linear systems. Prevalent and frequently used eigenvalue analysis based ROM technique in linear structural dynamics is the system equivalent reduction expansion process (SEREP) (O'Callahan, 1989).

In this study, SEREP has been explored in controlling the dynamics of a piezo-actuated flexible cantilever beam. The eigenvalue analysis of the full order system matrices is carried out to reduce the system dimension using SEREP. Subsequently, optimal linear quadratic regulator (LQR) is employed on the reduced system matrices, which have been obtained through SEREP. Traditionally, the gain matrix, to control the system dynamics, is obtained by solving full order dynamical systems, which is not computationally efficient. On the other hand, in the present analysis, this control gain matrix is obtained in reduced subspace which reduces the computational cost. The gain matrix obtained through SEREP is further used to estimate the required control forces for controlling the dynamics of the full order system.

## 4.2 Mathematical formulation of beam with PZT

The proposed philosophy of controlling dynamics of flexible structures integrating ROM is discussed in next section. The homogeneous, isotropic, thin Euler-Bernoulli beam with three collocated piezoelectric actuator and sensor pairs as shown in Fig. 4.1 is being solved using proposed algorithm. it is assumed that the piezoelectric layers are perfectly bonded with the host structure using adhesive. The adhesive layer is very thin compared to other dimensions of the structure and its contribution in the dynamics of the beam is neglected in developing the mathematical formulation. The dynamics of the beam is approximated using Galerkin projection method. This method leads to discretization of beam into smaller elements pertaining solving large degree of freedom system. To save the computational time the dimensionality of the problem is reduced using global model order reduction technique i.e., SEREP. The numerical simulation is performed on the system shown in Fig. 4.1 under various loading condition to check the efficacy of the proposed technique.

### 4.2.1 Linear cantilever beam

The previous chapter has developed equations of motion for structures with piezoelectric sensors and actuators. This sections builds up on that model to develop the equations of motion for a cantilever beam with piezoelectric sensor and actuator. As mentioned earlier any contribution of bonding agent between piezoelectric transducers and host structure in system dynamics are ignored. However, the contribution of piezoelectric actuators and sensors to the beam's mass and stiffness matrices has been considered in the formulation. The mathematical formulation for system collocated piezoelectric actuator and sensor pairs as shown in Fig. 4.1 is discussed here. The linear beam is modelled using the Euler-Bernoulli beam theory. The governing differential equation for transverse vibration of beam with piezoelectric sensor-actuator pair for this system

is given by using Eq. (3.40) as,

$$\frac{\partial^2 \left(M + \sum_{i=1}^{m} M_p^i \mathbb{H}_i(x)\right)}{\partial x^2} + \left(\rho A + \sum_{i=1}^{m} \rho_p^i A_p^i \mathbb{H}_i(x)\right) \frac{\partial^2 w}{\partial t^2} = F_{ext}(t) + U(x,t) \tag{4.1}$$

Where $U$ is the force due to the piezoelectric actuator, the subscript $(\cdot)_p^i$ refers to the properties of $i^{th}$ piezoelectric material. Here $i$ ranges from $1, 2, 3$. The sensor voltage

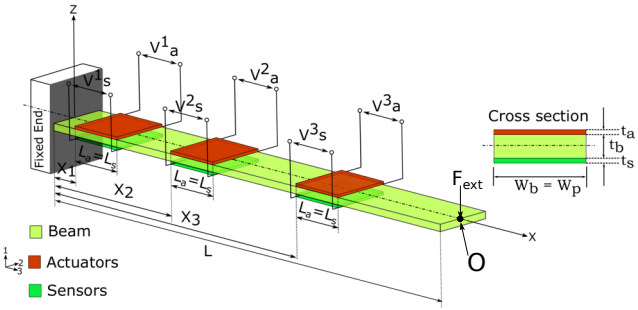



Fig. 4.1: Schematic of the cantilever beam with sensors and actuators in collocated configuration.

$\phi_s$ contributed by the bending effect can be estimated using the constitutive relation of the piezoelectric material in Eq. (3.37). When electric displacement $D = 0$, the electric field can be written in terms of strain as follow,

$$E = \frac{e_{31}}{\epsilon_{33}} \varepsilon_{xx} \tag{4.2}$$

$$\phi_s = \frac{t_s e_{31}}{\epsilon_{33}} z \frac{\partial^2 w}{\partial x^2} \tag{4.3}$$

The actuator voltage $\phi_a$ applied across the actuator can be determined using the sensing signal $\phi_s$ with control law discussed in section 4.4. The solution of Eq. (4.1) is numerically approximated using finite element formulation discussed in section 3.6. This leads

to set of second order differential equations similar to Eq. (3.66) and is given as,

$$\begin{bmatrix} M_{ww} & 0 \\ 0 & 0 \end{bmatrix} \begin{Bmatrix} \ddot{X} \\ \ddot{\varphi} \end{Bmatrix} + \begin{bmatrix} [K_{ww}] & [K_{w\varphi}] \\ [K_{\varphi w}] & [K_{\varphi\varphi}] \end{bmatrix} \begin{Bmatrix} X \\ \varphi \end{Bmatrix} = \begin{Bmatrix} F_{ext} \\ Q \end{Bmatrix} \tag{4.4}$$

In the distributed sensing and vibration suppression analysis, the displacement time history is much more important than the electrical potential vector in most applications. It is unnecessary to save the electrical potential vector at each time-domain integration unless the dynamic sensing time history is required. Thus, to save computer memory and improve computation efficiency, the electrical potential vector is usually condensed in the time domain integration.

$$\begin{bmatrix} [K_{ww}] & [K_{w\varphi}] \\ [K_{\varphi w}] & [K_{\varphi\varphi}] \end{bmatrix} \begin{Bmatrix} X \\ \varphi \end{Bmatrix} = \begin{Bmatrix} F_{ext} \\ Q \end{Bmatrix} \tag{4.5}$$

A congruent transformation matrix $[T_c]$ can be calculated as

$$[T_c] = \left[[I] - [K_{w\varphi}][K_{\varphi\varphi}]^{-1}\right] \tag{4.6}$$

$$[\mathbf{M}]\{\ddot{X}\} + [\mathbf{K}]\{X\} = \{\mathbf{F}_{ext}\} + \{\mathbf{F}_u\} \tag{4.7}$$

$$\{\mathbf{F}_u\} = -[K_{w\varphi}][K_{\varphi\varphi}]^{-1}\{Q\} \tag{4.8}$$

$$[\mathbf{K}] = [K_{ww}] - [K_{w\varphi}][K_{\varphi\varphi}]^{-1}[K_{\varphi w}] \tag{4.9}$$

In Eq. (4.7), there are two excitation forces associated with the piezoelectric structures: i.e., the mechanical forces and the electrical forces. The electrical potential vector can be recovered by,

$$\{\varphi\} = [K_{\varphi\varphi}]^{-1}\left(\{Q\} - [K_{\varphi w}]w\right) \tag{4.10}$$

The system dynamics is governed by equation Eq. (4.7) and the distributed dynamic measurement (voltage distribution) can be calculated by using equation Eq. (4.10). The

equation of motion for an undamped system in state-space is given as follow,

$$\mathbf{M}\ddot{X}(t) + \mathbf{K}X(t) = \mathbf{F}_{ext} + \mathbf{F}_u \tag{4.11}$$

Where, **M** is the global mass matrix, $X$ is the vector that consists of transverse displacement and rotation of the structural part of the system. Here, $F_{ext}$ is the externally applied load on the system, and $F_u$ is the control force applied by the actuators. Now considering the damping effect of structure (here, Rayleigh damping), the governing second order differential equation for the damped system in compact form is given by

$$\mathbf{M}\ddot{X}(t) + \mathbf{C}\dot{X}(t) + \mathbf{K}X(t) = \mathbf{F}_{ext} + \mathbf{F}_u \tag{4.12}$$

Here, **M**, **C** and **K** are of dimension $N \times N$. Moreover, $X(t), \dot{X}(t)$ and $\ddot{X}(t)$ represent the displacement, velocity and acceleration vector of dimension $N \times 1$, $\mathbf{F}_u$ is the force exerted by controller and $\mathbf{F}_{ext}$ is the externally applied force to the system. The mass and stiffness matrices of the piezoelectric material attached to the beam are incorporated in the global **M** and **K** matrices.

## 4.3 System Equivalent Reduction Expansion Process

System equivalent reduction expansion process (SEREP) is a global reduction algorithm that is based on the eigenvalue analysis of the high fidelity finite element model. It is a two-stage reduction process; reduction in a number of modes as well as the reduction in dofs. The transformation matrix obtained using this algorithm maps the full model to a lower subspace using selected eigenmodes with fewer arbitrary dofs. Eq. (4.12) describes the dynamics of the full order system and eigenvalue analysis using full order mass (**M**) and stiffness (**K**) matrices will give the modal matrix ($\psi \in \Re^{n\times n}$). Using this modal matrix, system can be written in terms of modal coordinates ($q(t) \in \Re^{n\times 1}$) as

$$X(t) = \psi_{n\times n} q_{n\times 1}(t) \tag{4.13}$$

Next, if only $a$ eigenmodes are retained, the solution can be written as

$$X(t) = \psi_{n\times a} q_{a\times 1}(t) \tag{4.14}$$

Now, splitting the full order system into master and slave dofs and retaining only $a$ eigenmodes in Eq. (4.14), which can be written in terms of active dofs as

$$X(t) = \begin{Bmatrix} X^m(t) \\ X^s(t) \end{Bmatrix} \approx \begin{Bmatrix} \psi^m \\ \psi^s \end{Bmatrix} q_a(t) \tag{4.15}$$

Where $q_a$ is the modal matrix having $a$ modes, the superscript $m$ and $s$ corresponds to master and slave, respectively. Also here, $\psi^m \in \Re^{m\times a}$ and $\psi^s \in \Re^{(n-m)\times a}$. Considering only the master dofs, the modal response in terms of master dofs can be given as

$$q_a(t) = \psi_p^m X^m(t) \tag{4.16}$$

where $\psi_p^m$ is the generalized inverse of $\psi^m$. The number of modes and number of dofs in the analysis may be the same, it depends on the problem, and hence the matrix $\psi^m$ always may not be a square. The different possible cases of the selection of modes and dofs have been discussed in (Lal *et al.*, 2017). Now substituting back Eq. (4.16) in Eq. (4.15), the full order response can be written as

$$X(t) = \psi\, \psi_p^m X^m(t) = T_r X^m(t) \tag{4.17}$$

Here, $\psi^m \psi_p^m$ is the transformation matrix ($T_r$) that maps the full system into reduced subspace. Using this transformation matrix, the full order system matrices in reduced dimension can be given as

$$\mathbf{M}_r = T_r^T \mathbf{M} T_r \qquad \mathbf{K}_r = T_r^T \mathbf{K} T_r \qquad \mathbf{C}_r = T_r^T \mathbf{C} T_r \tag{4.18}$$

where, $\mathbf{M}_r$, $\mathbf{K}_r$ and $\mathbf{C}_r$ represent the reduced mass, stiffness and damping matrices in $\Re^{m\times m}$. The superscript T in Eq. (4.18) denotes the transpose operator.

## 4.4 Control algorithm for deterministic systems

The dynamic equations for a cantilever beam with PZT in reduced dimension using SEREP can be expressed as,

$$\mathbf{M}_r\ddot{X}^m(t) + \mathbf{C}_r\dot{X}^m(t) + \mathbf{K}_r X^m(t) = \mathbf{F}_r(t) \tag{4.19}$$

For efficient handling of MIMO systems, Eq. (4.19) in the state space form can be

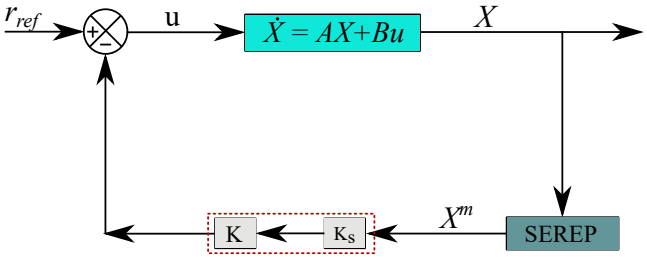



Fig. 4.2: Block diagram to show the flow of proposed control algorithm.

written as,

$$\dot{X}^m = A^m X^m + B^m u^m \tag{4.20}$$

where,

$$A^m = \begin{bmatrix} 0 & I \\ -\mathbf{M}_r^{-1}\mathbf{K}_r & -\mathbf{M}_r^{-1}\mathbf{C}_r \end{bmatrix}, \qquad B^m = \begin{bmatrix} 0 \\ -\mathbf{M}_r^{-1}\mathbf{F}_r \end{bmatrix} \tag{4.21}$$

where, $\dot{X}^m$ is $(m \times 1)$ state vector. In this study, a Linear Quadratic Regulator (LQR) controller is used which has been implemented on the proposed reduced order model algorithm. The optimal control input to the actuator for vibration suppression is obtained through LQR, minimising the cost function. This cost function $(J)$ is defined as,

$$J = \lim_{\tau\to\infty} \frac{1}{\tau} \left[ \int_0^\infty \{(X^m)^T Q(X^m) + (u^m)^T R(u^m)\} dt \right] \tag{4.22}$$

Where $Q$ and $R$ are weighting matrices which penalize the system states $X^m$ and control input $u^m$. Minimization of the quadratic cost function defined in Eq. (4.22) subjected to system dynamics, Eq. (4.20) gives the state feedback in form of required control input $u^m$ which is given by,

$$u^m = -K_s X^m \tag{4.23}$$

Where $K_s$ is the $(m \times r)$ gain matrix obtained through the proposed reduced-order model. The closed-loop optimal gain is calculated by solving the reduced-matrix Riccati Equation. Using this gain matrix and transformation matrix $T_r$ the control input needed to control the full order system dynamics is estimated. The block diagram of the proposed framework is shown in Fig. 2. Here, in this analysis, the value of $r_{ref}$ has been assumed to be zero.

## 4.5 Numerical example

An example of flexible cantilever beam length with piezoelectric patches, as shown in Fig. 4.5 has been solved numerically to verify the efficiency of the proposed reduce order formulation. The beam's geometric dimension material properties have been taken as shown in table 4.1. Fig. 4.3 shows the convergence study of the number of elements in the finite discretization of the system. It can be seen that after 200 elements, the error norm of natural frequencies has become constant. Hence, based on the convergence study, the system has been discretized using 200 elements in the current analysis. Each element has two nodes, and each node has two dofs. To discuss the importance of selection of nodes in the formulation of SEREP, two different cases have been considered.

- Case 1: Nodes are selected at an equal interval in the mesh, as shown in Fig. 4.4. Here, in this way of selection of nodes, some of the critical points have been missed. In this case, around 48 nodes have been chosen, and hence a total of 96 dofs will participate in the analysis.
- Case 2: Nodes are selected systematically from the entire domain as reported in Fig. 4.5. Here, all the critical nodes has been taken into consideration and only 19 nodes has been chosen and hence a total of 38 dofs.

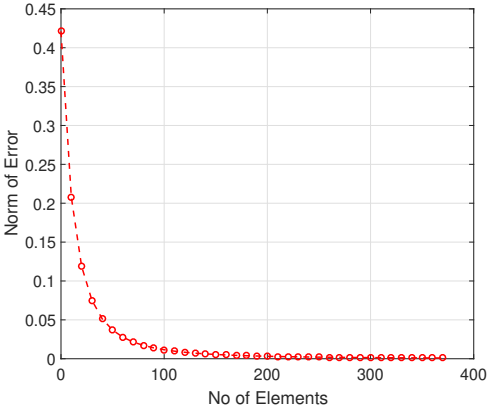



Fig. 4.3: Convergence of number of elements for finite element analysis.

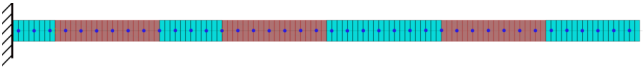


Fig. 4.4: Location of selected nodes for analysis of system using SEREP in Case 1.

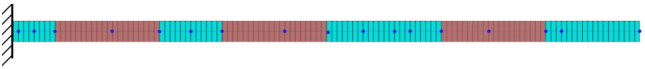


Fig. 4.5: Location of selected nodes for analysis of system using SEREP in Case 2.

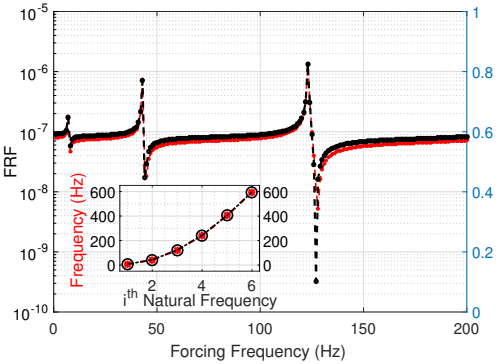



Fig. 4.6: Comparison of frequency response obtained through full order FE analysis and SEREP and comparison of first five natural frequencies of the system is shown in inset (for Case 2).

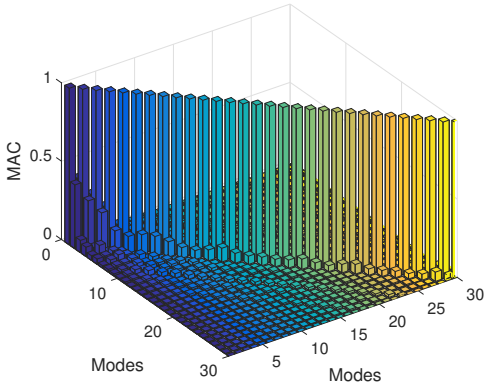
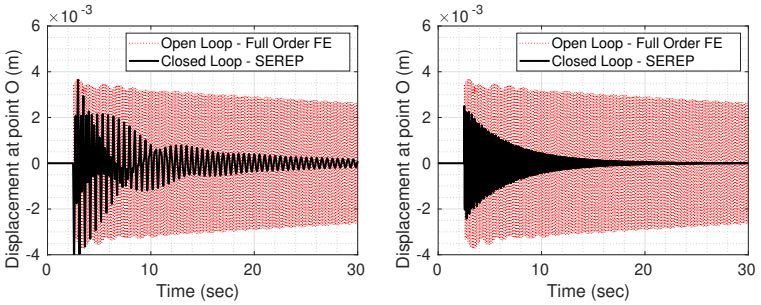



Fig. 4.7: Modal assurance criteria of modes obtained through SEREP with the modes of full order system.

Fig. 4.6 shows the comparison of the full order system response to that obtained through SEREP. In the inset of Fig. 4.6, the comparison of the first five natural frequencies has also been depicted. It can be noticed here that the results obtained through SEREP are in very good agreement with those obtained through the full-order system. To show the comparison of mode shapes obtained through SEREP with that of full order system, modal assurance criteria (MAC) has been shown in Fig. 4.7. The definition of MAC is given in (Ewins, 2009) Figure 4.9 shows a comparison of the velocity profile

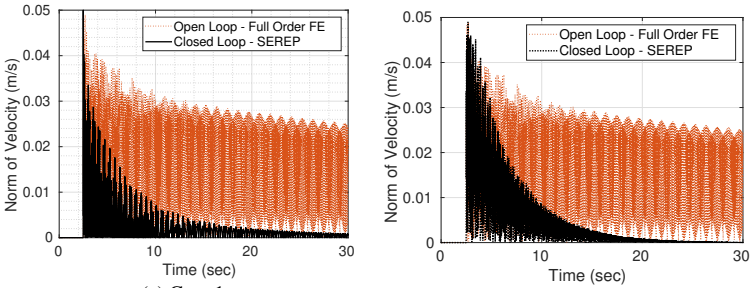



(a) Case 1. (b) Case 2.

Fig. 4.8: Comparison of controlled response with that of the uncontrolled dynamics

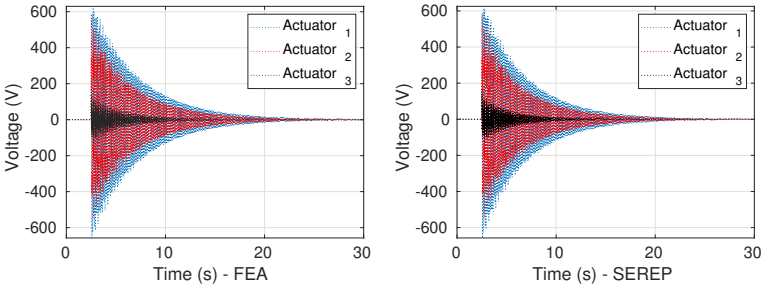
of the beam for open loop and closed loop in both the cases. In this figure the norm of

the velocity is plotted with respect to the time which is a measure of the kinetic energy in the system. Kinetic energy of the controlled system in the both case is keep reducing as time progresses. The energy of system in case 1 is much more as shown in Fig. 4.9a compared to case 2 as shown in Fig. 4.9b after 20 seconds. It shows that the system is better controlled in case 2 compared to case 1. This happens because the choice of DoFs. Moreover, controlled response from Case 2 shows very good agreement with that of full order FE system as shown in Fig. 4.11.




(a) Case 1. (b) Case 2.

Fig. 4.9: Comparison of controlled Norm of velocity with that of the uncontrolled response




(a) Case 2. (b) Case 2.

Fig. 4.10: Required voltage needed given to the actuators to control the dynamics

$$MAC_i = \frac{|v_{ir}^T v_{if}|^2}{(v_{ir}^T v_{ir})(v_{if}^T v_{if})} \tag{4.24}$$

Where, $v_{ir}$ and $v_{if}$ are the $i^{th}$ normal modes computed using the reduced-order modal

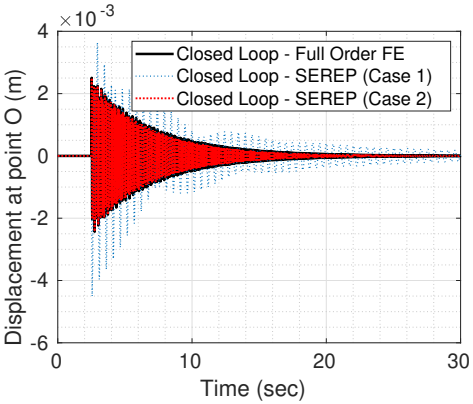



Fig. 4.11: Comparison of controlled response obtained through full order FE model and proposed algorithm.

and the full order system, respectively. MAC value of one represents the best match of modes obtained through ROM to that with the full order modes. In Fig. 4.7, MAC value for the first 30 modes has been shown, which are very close to one and hence the modes of full order system has been retained in the reduced space. The controlled response obtained at free end (Point 'O') through ROM using the set of selected dofs as mentioned in Case 1 has been shown in Fig. 4.8a. The dynamic response of the system is getting reduced but not going to zero. The reason behind that is during the selection of master degree of freedoms critical points were being ignored in this case. Hence, the control response obtained from full order also shows disagreement with that of obtained from Case 1. The amount of required voltage given to the actuators for full order FE analysis has been reported in Fig. 4.10a. On the other hand, with less number of dofs in SEREP formulation (Case 2), the controlled dynamic response is almost going to zero after 20 seconds, as shown in Fig. 4.8b. The corresponding voltage required for controlling the uncontrolled response has been shown in Fig. 4.10b.

The controlled responses obtained through SEREP (Case 1 and Case 2) have been compared with the controlled response obtained with the help of gain matrix using the full order system matrices as shown in Fig. 4.11. In case 1, the way nodes are selected (at equal interval), the nodes corresponding to junctions of actuators and beam are missing in the formulation. Whereas in the other case, only 38 dofs has been chosen

Table 4.1: Material properties of beam with piezoelectric patches.

| Property | Value | Units |
|---|---|---|
| Length of the beam | $0.5$ | $m$ |
| Width of the beam | $0.03$ | $m$ |
| Thickness of the beam | $0.002$ | $m$ |
| Elastic modulus of beam | $210$ | $GPa$ |
| Density of beam | $7800$ | $kg/m^3$ |
| poisson's ratio | $0.3$ | |
| Elastic modulus of piezo transducers | $139$ | $GPa$ |
| Density of piezo transducers | $7500$ | $kg/m^3$ |
| Piezoelectric strain coefficient for sensor ($d_{31}$) | $-250 \times 10^{-12}$ | $C/N$ |
| Piezoelectric strain coefficient for actuator ($d_{33}$) | $650 \times 10^{-12}$ | $C/N$ |

systematically from the entire domain. Out of 38 dofs, 16 dofs are chosen at critical points and remaining from entire domain. Therefore, even though the number of dofs in Case 1 is more than that in Case 2, the response is getting better controlled in the later case as can be seen in Fig. 4.11.

The full order finite element of the system has a total of 200 elements which results in a sum of 402 dofs and consequently the state dimension will be 804. Here, in SEREP framework, two cases have been considered: one with 96 dofs and another case with 38 dofs. The reduction in computational cost for first case has been achieved by 95% whereas for the other case this value has been increase to 97.5%. Even though the reduction in a number of dofs in Case 2 is approximately 60% relative to Case 1, the global improvement in computational cost is around 2.5%. Essentially, SEREP is based on the eigenvalue analysis of full system matrices to get the transformation matrix. This eigenvalue analysis is a costly process and an inevitable step in this framework, which has been reflected in the difference among the two aforementioned cases' computational cost.

### 4.5.1 System under periodic loading

The efficacy of developed methodology is validated again using another example. The cantilever beam, as shown in Fig. 4.1 has been subjected to periodic excitation force as shown in Fig. 4.12 at free end (O). The beam's geometric dimension and properties are same as of the previous example. The selection of dofs are done in such a way that

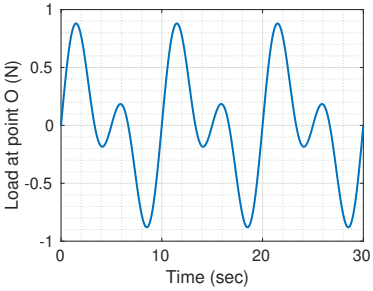
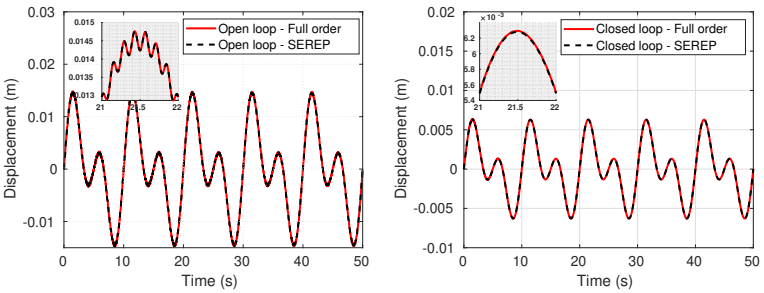


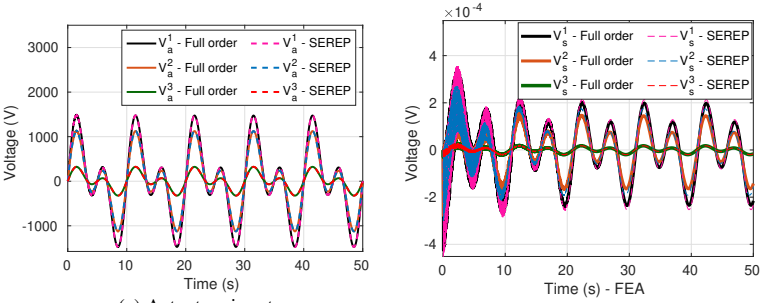

Fig. 4.12: Force applied at free end of beam




(a) Uncontrolled response (b) Controlled response

Fig. 4.13: Comparison of dynamics of end node obtained through full order FE model and proposed algorithm.

the dynamics of full order system is captured accurately. The uncontrolled response in presence of external harmonic excitation obtained through full order and ROM using the set of selected dofs as mentioned in Case 2 has been shown in Fig. 4.13a. The controlled dynamic response of the system in both the cases is shown in Fig. 4.13b. From comparing insets of Fig. 4.13a and Fig. 4.13b it is seen that the amplitude of both transient, steady state dynamics of system is reduced and the dynamics from the proposed algorithm shows very good agreement to that of Full order finite element method. The voltage produced by sensors is shown in Fig. 4.14b and amount of required voltage given to the actuators has been reported in Fig. 4.14a. In this example full order finite element of the system has a total of 200 elements which results in a sum of 402 dofs and consequently the state dimension will be 804. Here, in SEREP framework, the




(a) Actuators input

(b) Sensors output

Fig. 4.14: Comparison of piezoelectric transducers response obtained through full order FE model and proposed algorithm.

case 2 from previous example is considered i.e., case with 38 dofs. The reduction in computational cost for this example has been achieved by 96.1% compared to full order finite element simulation.

## 4.6 Summary

This chapter describes the full order FE modelling of a linear deterministic flexible cantilever beam with piezoelectric actuators. The philosophy and procedure to obtain the transformation matrix, which maps the full system to reduced dimension using SEREP, is also explained in this chapter. Followed by the procedure to apply the control algorithm to calculate the gain matrix and subsequently estimating the control force in reduced subspace. The chapter concludes with the findings through the solution of a benchmark problem of cantilever beam.

# CHAPTER 5

# Reduced order model integrated control in presence of parametric uncertainty

From Chapter 4 it can be seen that it is possible to control the dynamics of linear deterministic structures using control developed over a reduced order model. But in real world it is impossible to make structures that is deterministic in true sense. This chapter address the issue of controlling such systems. The chapter starts with a small introduction to this problem followed by mathematical description. Numerical example of cantilever beam with varying density and modulus of elasticity is solved using this proposed algorithm and results are reported in the end of the chapter.

## 5.1 Introduction

Parametric uncertainties are inevitable in all stages of ranging from manufacturing, process design and synthesis, to operational planning, production scheduling and supply chain optimization. Uncertainties associated with structural characteristics introduces variability in the dynamic response structures. The system parameters show stochastic behaviour because of these manufacturing limitations or/and measurement errors. The predicted dynamical behavior of these structures based on mathematical models with imperfect knowledge of their material characteristics tends to be different from reality. The system parameters are either not known precisely or are expected to vary in the operational lifetime (Ramachandran *et al.*, 2003). In either case, the distribution of variation of these parameters can be measured or calculated. It is desired to design a controller for such systems considering parametric uncertainty. This will result in the amplitude of vibration to remain within the acceptable limits.

Many control algorithms are developed for structures with parametric uncertainty (Fan and Tits, 1991; Zhou and Doyle, 1998). In-state space, robust control design methods

address parametric uncertainty as structured uncertainty, modelled using linear fractional transformation (LFT) (John, 1982). Worst-case analysis is performed on aforementioned methods to determine the effect of parameter variations on the robustness of the control algorithm. Thus, these methods can be conservative if the parameters rarely assume the extreme values. Hence, there is a need for a control technique that exploits the probability distribution of parameters. It requires the knowledge of the probability distribution function (PDF) of the system parameters, which will stabilize the system for all possible parameters with a non-zero probability of occurrence and optimize the performance with respect to the PDF of the parameters (Housner *et al.*, 1997). Many techniques for stability analysis and controller design for stochastic systems are reported in literature (Yongbo Peng, 2019). It reports the evolution of state trajectories when deterministic dynamical systems are subjected to stochastic forcing and analysis is restricted to characterization of mean trajectory. In (Skelton *et al.*, 1997), the problem of covariance control with Gaussian excitation is investigated.

Firstly, a methodology needs to be developed to incorporate these parametric uncertainties into the computational mathematical model. Here, the starting point is the weak form representation using the flexible structures' FE model, which are usually modelled as continuous system through coupled partial differential equations. Now, the weak form representation contains the coupled differential equations in terms of nodal variables. As mentioned, the full-order FE model introduces a larger number of nodal variables into the mathematical model and, consequently, increases the problem's dimension, hence the associated computational cost. To bypass the complexity corresponding, reduced-order models (ROMs) are developed. Since model reduction technique in a control system is an active area of research (James Fisher, 2009), the problem considered in this study take into account the additional complexity that arises due to the uncertainty associated with system parameters.

The weak form representation of a stochastically parametered system consists of coupled differential equations whose coefficients are functions of vector of random variables $\zeta$. The propagation of these input uncertainties needs to be characterized in the probabilistic sense. In the current study, SEREP is used for reducing the state space dimension of the system, which involves the solution of the random eigenvalue prob-

lem. Further, the control algorithm is developed in conjunction with SEREP, where the optimal linear quadratic regulator is employed on reduced system matrices. The control gain matrix is also evaluated in reduced subspace, contrary to the traditional approach where full order matrices are used. For probabilistic characterization, Monte Carlo (MC) simulation approach is used, which involves the repeated numerical simulation of the deterministic system for various realizations of $\boldsymbol{\zeta}$.

## 5.2 Mathematical formulation for linear stochastic cantilever beam

For dynamical systems with stochastic parametric uncertainties, structural parameters are assumed to be random variables and to follow non-Gaussian distributions. The random variables are collectively represented as $\boldsymbol{\zeta}(\mathbb{w})$, defined in the probability space $(\mathbb{\Omega}, \mathbb{A}, \mathbb{P})$, $\mathbb{w} \in \mathbb{\Omega}$. The corresponding governing equation for a stochastically parameterized dynamical system is given as,

$$\mathbf{M}\ddot{X}(t;\boldsymbol{\zeta}(\mathbb{w})) + \mathbf{C}(\boldsymbol{\zeta}(\mathbb{w}))\dot{X}(t;\boldsymbol{\zeta}(\mathbb{w})) + \mathbf{K}(\boldsymbol{\zeta}(\mathbb{w}))X(t;\boldsymbol{\zeta}(\mathbb{w})) = \mathbf{F}_u + \mathbf{F}_{ext} \tag{5.1}$$

Here, the force vector is assumed to be deterministic in full order analysis. The same transformation matrix is used for reducing the force vector in a reduced domain by $T_r^T F(t)$. Using SEREP, the modes of full order dynamical systems are also retained in the reduced subspace. To retrieve the full order system dynamics, the transformation matrix ($T(\mathbb{W})$) is used. The detail procedure to obtain the transformation matrix in deterministic sense is explained in the previous chapter. Moreover, selection of modes and dofs are governed by the type of system as well as the external loading conditions are retrieved from $T(\mathbb{W})$. However, the selection of active dofs and modes in the formulation from an entire domain is preferable. Extending the deterministic formulation obtained in previous chapter for the stochastic case, Eq. (4.17) can be written as,

$$X(t;\mathbb{w}) = \mathbf{T}(\mathbb{w})X_a(t;\mathbb{w}) \tag{5.2}$$

where, the transformation matrix $\mathbf{T}(\mathbb{w}) = \mathbf{\Psi}(\mathbb{w})\,\mathbf{\Psi}_a^{\dagger}(\mathbb{w})$ is also stochastic. The corresponding stochastic system of equation in reduced dimension will be written as,

$$\mathbf{M}_r\ddot{X}(t;\mathbb{w}) + \mathbf{C}_r(\mathbb{w})\dot{X}(t;\mathbb{w}) + \mathbf{K}_r(\mathbb{w})X(t;\mathbb{w}) = \mathbf{F}_r(t;\mathbb{w}) \tag{5.3}$$

Here, $\mathbf{M}_r = \mathbf{T}^T(\mathbb{w})\mathbf{M}\mathbf{T}(\mathbb{w})$, $\mathbf{C}_r(\mathbb{w}) = \mathbf{T}^T(\mathbb{w})\mathbf{C}(\mathbb{w})\mathbf{T}(\mathbb{w})$, $\mathbf{K}_r(\mathbb{w}) = \mathbf{T}^T(\mathbb{w})\mathbf{K}(\mathbb{w})\mathbf{T}(\mathbb{w})$ and $\mathbf{F}_r(t;\mathbb{w}) = \mathbf{T}^T(\mathbb{w})\mathbf{F}(t)$ are reduced mass, damping, stiffness metrics and force vector, where $\mathbf{F}(t) = \mathbf{F}_u + \mathbf{F}_{ext}(t)$. It can be noted here that even though the force vector is deterministic in full order model, it is stochastic in reduced subspace because of the transformation matrix being stochastic.

## 5.3 Control algorithm of stochastic system

The dynamic Eq. (5.3) for stochastic cantilever beam with PZTs in reduced dimensions using SEREP is represented in control canonical form (Zhou and Doyle, 1998) as,

$$\mathbf{X}(t;\mathbb{w}) = A(\mathbb{w})\mathbf{X}(t;\mathbb{w}) + B(\mathbb{w})U(t;\mathbb{w}) \tag{5.4}$$

In this study, a deterministic Linear Quadratic Regulator (LQR) is used to obtain control. The uncertainty introduced in the system is a random variable type, and the system dynamics are solved using Monte-Carlo simulation, which means the system is solved for each realization. If the system is solved for each realization with keeping in mind the uncertainty is a random variable type, a deterministic controller can be used (James Fisher, 2009) to obtain the controller gain for each realization. The reduced-order dynamical Eq. (5.4) for $i^{th}$ realization is subjected to standard quadratic cost function given as,

$$J_i(\mathbb{w}) = \lim_{\tau\to\infty}\frac{1}{\tau}\left[\int_0^{\infty}\{(\mathbf{X}_i(t;\mathbb{w}))^T Q(\mathbf{X}_i(t;\mathbb{w})) + (U_i(t;\mathbb{w}))^T R(U_i(t;\mathbb{w}))\}dt\right] \tag{5.5}$$

Where $Q$ and $R$ are weighting matrices that penalize the system states and control input. Minimization of the quadratic cost function $J_i$ subjected to system dynamics in

each realization gives the control input in state-feedback form given by,

$$U_i(t;\mathbb{w}) = -R^{-1}B_{p_i}^T P_i \mathbf{X}_i(t;\mathbb{w}) \tag{5.6}$$

$$U_i(t;\mathbb{w}) = -\mathbf{K}_i(t;\mathbb{w})\mathbf{X}_i(t;\mathbb{w}) \tag{5.7}$$

where, $P_i$ is solution to Matrix Riccati (James Fisher, 2009) equation for $i^{th}$ realization. The control gain matrix is obtained through the proposed reduced order model and input needed for to control the dynamics of full order system is estimated.

## 5.4 Numerical example

A flexible cantilever beam with piezoelectric actuators and sensors in collocated configuration, as shown in Fig. 4.1, is considered to illustrate the efficiency of the proposed framework. The geometric dimensions and material properties of the beam are taken from Table 4.1. The system's FE model is developed with two noded linear beam elements comprising 200 elements, where each node has two dofs. The dofs associated with the left boundary node are taken to be fixed boundary conditions. The elastic modulus is modelled as a lognormally distributed random variable, with the mean value of 210 GPa. The SEREP based ROM of the system is obtained through an eigenvalue analysis on the deterministic system. The selection of dofs for reduced-order model is arbitrary and should be in such a way so that it preserves the dynamics of the full order system. However, there are few guidelines that help in the selection of the critical dofs for SEREP. For example, the dofs correspond to the node where structural geometric properties are varying or external loads are applied, which need to be considered. The dofs are chosen which represents the dynamics of full order system, are called active dofs. The effect of active dofs on the estimation of responses using proposed framework are reported in Table 5.1 by calculating the mean square error (MSE) of the response and the corresponding computational costs. It can be noticed here that the MSE for controlled dynamics is more compared to the uncontrolled system for the same number of active dofs because the placement of actuators and sensors poses additional complexity and more critical locations to the system. Also, the dofs corresponding to the actuator

location have a negligible effect on the system response obtained through ROM for the uncontrolled system. On the other hand,

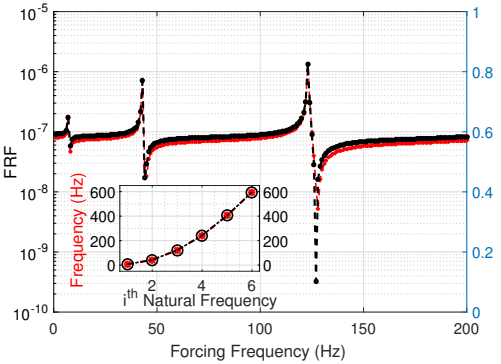



Fig. 5.1: Comparison of FRF and natural frequencies obtained through SEREP and full order FE analysis.

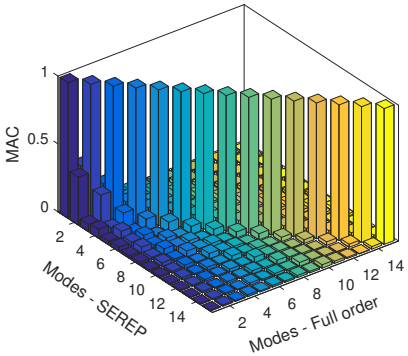
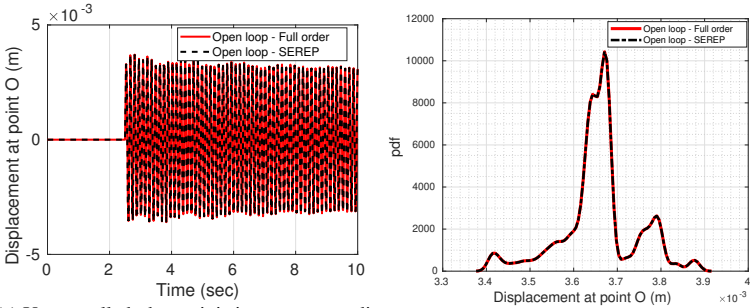



Fig. 5.2: Modal Assurance Criterion to compare the modes obtained through SEREP and full order FE analysis.

While solving the system in a closed-loop actuator effort makes the system stabilized, the corresponding dofs become critical and need to be considered. As the number of active dofs increases by selecting the more critical nodes, the error reduces and consequently, the computational cost goes up. There is always a trade-off between the error in

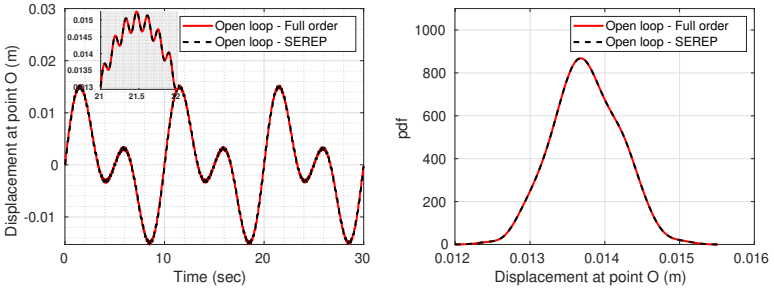



(a) Uncontrolled deterministic transverse displacement

(b) pdf of uncontrolled maximum transverse displacement

Fig. 5.3: Comparison of the transverse displacement under impulse loading (at the free end) obtained through full order FE analysis and SEREP.

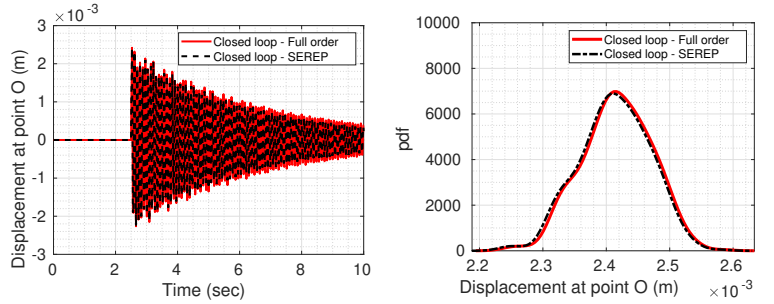
approximate solution and computational cost which is decided by the permissible error for the problem at hand. Next, the frequency response function (FRF) obtained through ROM is compared with that obtained through full order analysis as shown in Fig. 5.1. In the inset, first six natural frequencies of the structure obtained through SEREP are also compared with the full order results. The FRF and natural frequencies obtained through ROM are in good agreement with those obtained through the full order analysis. Here, for SEREP, a total of 320 active dofs are selected to reduce the state space dimension. To show the resemblance in mode shapes obtained through SEREP based ROM with that of full order system, Modal Assurance Criterion (MAC) (M. Brehm, 2010) is shown in Fig. 5.2. Here, MAC is obtained for the first fifteen mode shapes, and it is close to one which represents a very good match. Further, Fig. 5.3a shows the comparison of deterministic uncontrolled transverse displacement under impulse loading at the free end and Fig. 5.4a shows same under harmonic loading obtained through ROM and full order analysis when system parameters are taken to be the mean value. Here, results obtained through ROM and full order FE analysis show a very good match with each other. Now, stochastic analysis is carried out, for which 1000 realizations of a random variable are generated, and a random system of equations is solved using Monte Carlo simulation iteratively.




(a) Uncontrolled deterministic transverse displacement

(b) pdf of uncontrolled maximum transverse displacement

Fig. 5.4: Comparison of the transverse displacement under harmonic loading (at the free end) obtained through full order FE analysis and SEREP.



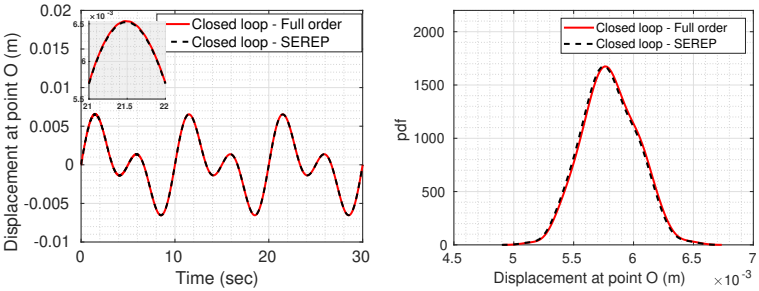

(a) Controlled deterministic transverse displacement

(b) pdf of controlled maximum transverse displacement

Fig. 5.5: Comparison of the transverse displacement under impulse loading (at the free end) obtained through full order FE analysis and SEREP.

Here, the dimension of system matrices will be equal to the number of active dofs and in each iteration system of equations will be solved in the reduced subspace. Further, using the same transformation matrix (**T**), the response quantities at each node will be evaluated. For the cantilever beam, free end undergoes maximum transverse displacement and need to control. Fig. 5.3b and Fig. 5.4b shows the comparison of the pdf of the uncontrolled transverse displacement at the node under impulse and harmonic loading corresponding to free end obtained through ROM and full order FE analysis. The value of 'pdf' on Y-axis scales according to the values on X-axis because area un-

der the pdf response always equal to the one. Here, the pdf of the response in both the cases show a very good agreement with each other. Next, the system dynamics is controlled using the proposed methodology, which includes the calculation of control gain in the reduced subspace.

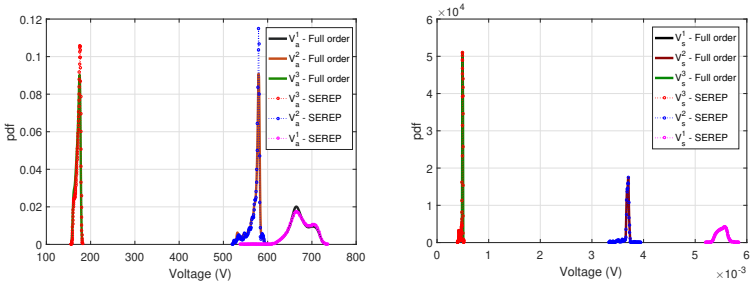



(a) Controlled deterministic transverse displacement

(b) pdf of controlled maximum transverse displacement

Fig. 5.6: Comparison of the transverse displacement under harmonic loading (at the free end) obtained through full order FE analysis and SEREP.

Fig. 5.5a and Fig. 5.6a shows one of the ensembles of the controlled dynamics at the same node obtained through SEREP and full order FE analysis for both loading conditions. It can be seen that the amplitude of vibration after 10 seconds is reduced by approximately 90 $\%$ after implementing the control algorithm in case of impulse loading. Next, the pdf of the system's maximum displacement under impulse loading condition integrated with a controller is shown in Fig. 5.5b and same for harmonic loading is shown in Fig. 5.6b. Here, the mean value of the maximum displacement in controlled dynamics is reduced by approximately 35 $\%$ compared to that of the uncontrolled system. In Fig. 5.5b, pdf of maximum response, at the free end of the beam, obtained through full order analysis and SEREP is compared. The pdf obtained through SEREP is showing a very close resemblance with that obtained through full order analysis. There are three actuators placed at different locations across the span of the beam to control the dynamics of the system, which provide the required amount of actuation to control the amplitude of vibration. Fig. 5.7a and Fig. 5.8a shows the pdf of the actuation input required to control the system vibration for all the actuators under impulse



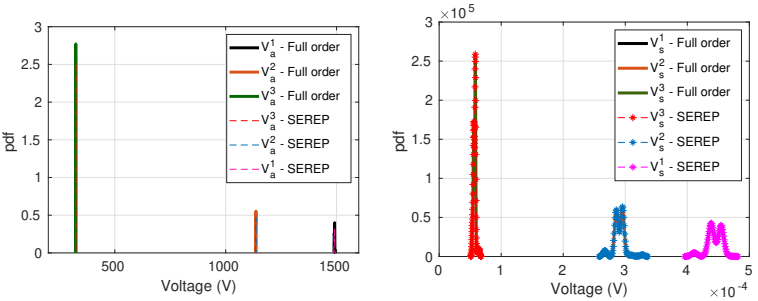

(a) pdf of actuators input (b) pdf of sensor output

Fig. 5.7: Comparison of pdf response through full order analysis and SEREP based ROM under impulse loading at free end.

Table 5.1: Comparison of norm of response and computational costs for different set of active dofs.

| Active dofs | $L^2$ norm | Time saved (%) |
|---|---|---|
| 50 | 0.9765 | 97.54 |
| 100 | 0.8965 | 94.36 |
| 200 | 0.8263 | 78.56 |
| 320 | 0.5632 | 45.60 |

and harmonic loading respectively.




(a) pdf of actuators input (b) pdf of sensors output

Fig. 5.8: Comparison of pdf response through full order analysis and SEREP based ROM under harmonic loading at free end.

The $L^2$ norm of controlled response in reduced order framework is compared with that of uncontrolled norm of FE response and reported in Table 5.1. The computational effort has been reduced drastically with the acceptable error tolerance as shown

in Table 5.1. On the other hand, Fig. 5.7b and Fig. 5.8b shows the voltage output of the sensors for both loading conditions, which is calculated using Eq. (4.10) for full order system and compared with that obtained through ROM and a very good match is obtained.

## 5.5 Summary

This chapter begins with describing the full order FE modeling of flexible cantilever beam with piezoelectric actuator and sensors. Moreover underlying principle of the reduced order model (SEREP) for stochastically parametered system is explained. Followed by, Control algorithm for stochastic reduced order model is derived in. The proposed mathematical framework is illustrated through a numerical example of stochastic cantilever beam and findings are reported at the end of the chapter.

# CHAPTER 6

# Summary, conclusion and future scope

## 6.1 Summary

Vibration control of deterministic and stochastic linear flexible cantilever beam with piezoelectric transducers using reduced order modelling is discussed in this thesis. Background and objectives of the thesis were drafted in Chapter 1. The historical development in control of the distributed parameter systems were documented as literature review Chapter 2. Various reduced order modelling techniques were also mentioned in the chapter. Piezoelectric actuators and sensors have been used in this thesis because of their excellent actuation and sensing abilities. Structures which lend themselves to piezoelectric transducer for vibration control application are flexible in nature. These structures are governed by PDEs and solution to such systems lies in infinite dimensional space. The solution to these systems are numerically approximated by the FE modelling. The complete modelling procedure of these smart structures were derived in Chapter 3. The developed equations of motion are solved in MATLAB under periodic excitation. To validate the numerical solution, same system has been constructed and solved in ABAQUS. The dynamics response of the end node in both case was compared along with the sensor output. The dynamical response from both software showed very good agreement. The full fidelity solution takes very long time to solve for system dynamics. To cater that the philosophy and procedure for developing reduced order model was discussed in Chapter 4. The detail mathematical procedure to obtain the transformation matrix, which maps the full system to reduced dimensions using SEREP, is also explained in the same chapter. This was followed by the procedure to apply LQR to estimate the required amount of control force for vibration control. Numerical simulation of deterministic linear flexible cantilever beam was carried out and the results were discussed. Moreover the developed philosophy was extended to the linear beam having parametric uncertainties. The underlying principle of reduced order model (SEREP) for

stochastically parametered system was explained in the Chapter 5. The proposed mathematical framework was illustrated through a numerical example of stochastic cantilever beam and dynamics of system was solved using Monte Carlo simulation. The chapter concluded with the findings and results were also discussed. The concluding remarks on the present work is discussed in the next section.

## 6.2 Conclusion

In this study, a reduced order model has been proposed to control the dynamics of a flexible structures. The conclusions drawn based on the study are as follows:

- The reduced model using SEREP provides good approximation to the full order system with both deterministic and uncertain parameters. It is also observed that the SEREP based RoM can capture the dynamics of beam, integrated with sensors and actuators.
- Control gain matrix obtained in reduced framework then transformed to the full order and feedback to full order system, provides efficient control of the beam system.
- The reduction in computational cost for the first case has been achieved by 95 $\%$ whereas for the other case this value has been increased to 97.5 $\%$.
- The pdf of controlled response for beam with uncertain parameters obtained from full order FE shows very good agreement with that of proposed framework with around 45.60 $\%$ of reduction in computational cost.

## 6.3 Future scope

The mathematical modelling of active stochastic control is a very challenging task. We attempt to study the basic mathematical modelling to check the feasibility of the proposed algorithm. Following are the list of the area that can be considered as a further study.

- Active vibration and shape control of nonlinear flexible structures can be considered as a further direction. This will help in understanding the morning dynamics quickly. Moreover, aircraft wings are twisted, tappered beam. One needs to check whether current SEREP model can determine the dynamics of these beams with reasonable accuracy and less number of DoF.

- The stochastic control algorithm can be developed by using polynomial chaos expansion. The beam can be modelled in a stochastic space with non-Gaussian random field considering material and geometric uncertainties. SEREP should be extended to such models in the stochastic space.
- Experimental studies can be taken up to validate the proposed mathematical modelling.

Finally, this chapter summarizes thesis, draws conclusion and provides areas of future direction based on the thesis contribution.

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